

What Can Economic Theory Say About AI?

Annie Liang
(Northwestern)

EC 2026

What does AI do for economic theory?

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Science fiction starts from the known world, changes one thing, and traces how its consequences propagate through society.

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AI substantially expands the set of plausible worlds

- new agents—nonhuman, human/AI hybrids, digital replicas
- new capabilities—cognition and creativity at scale
- new organizations—blurred distinction between capital and human capital

What does economic theory do for AI?

Economic theory can help us think through these changes by providing:

- **Normative frameworks:** how to evaluate welfare, and how to design the mechanisms, markets, and policies AI operates in
- **Extrapolation:** data describes worlds that exist; theory reasons about worlds that don't (yet)

Don't take it from me...



Dario Amodei, CEO of Anthropic

“Let’s just imagine we’re in an **economics textbook**. We have a small number of firms. Each can invest a limited amount...Companies will compete to push their research budgets up. But because there’s a small number of players, we have the... What is it called? The **Cournot equilibrium**...” (*Dwarkesh Podcast, 2026*)

“The **game theory** is a little tough to think through.” (*Dwarkesh Podcast, 2024*)



John Schulman, co-creator of ChatGPT

This Talk

1. A quick overview of questions about AI that seem especially suited to microeconomic theory/econ-CS
2. Some recent papers I have written inspired by these questions

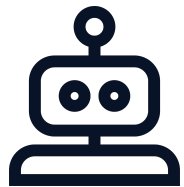
Part I

Questions

Levels of Resolution

Level 1:

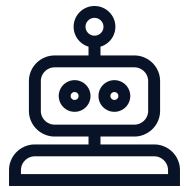
The AI itself



Levels of Resolution

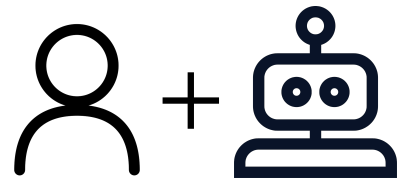
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The AI itself



Level 2:

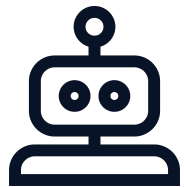
The Human+AI unit



Levels of Resolution

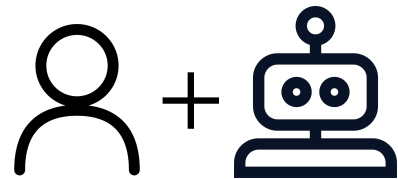
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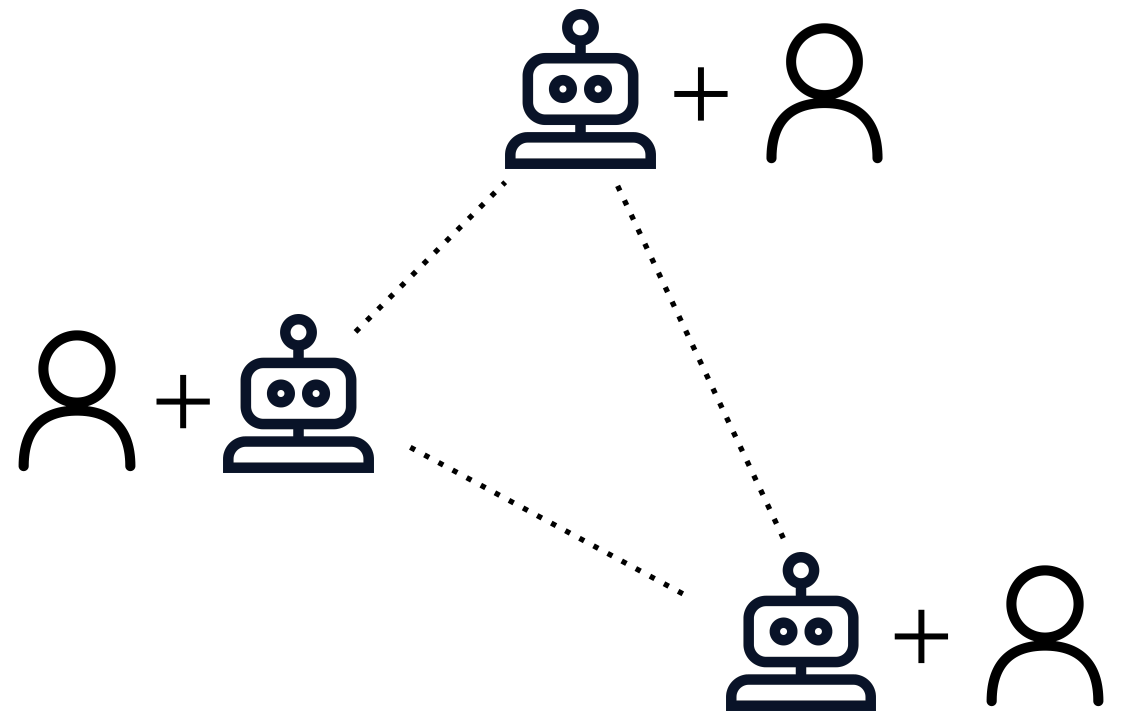
Level 2:

The Human+AI unit



Level 3:

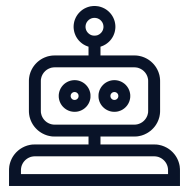
A society/market of
human+AI units



Levels of Resolution

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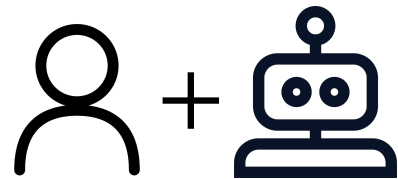
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Decision theory

Level 2:

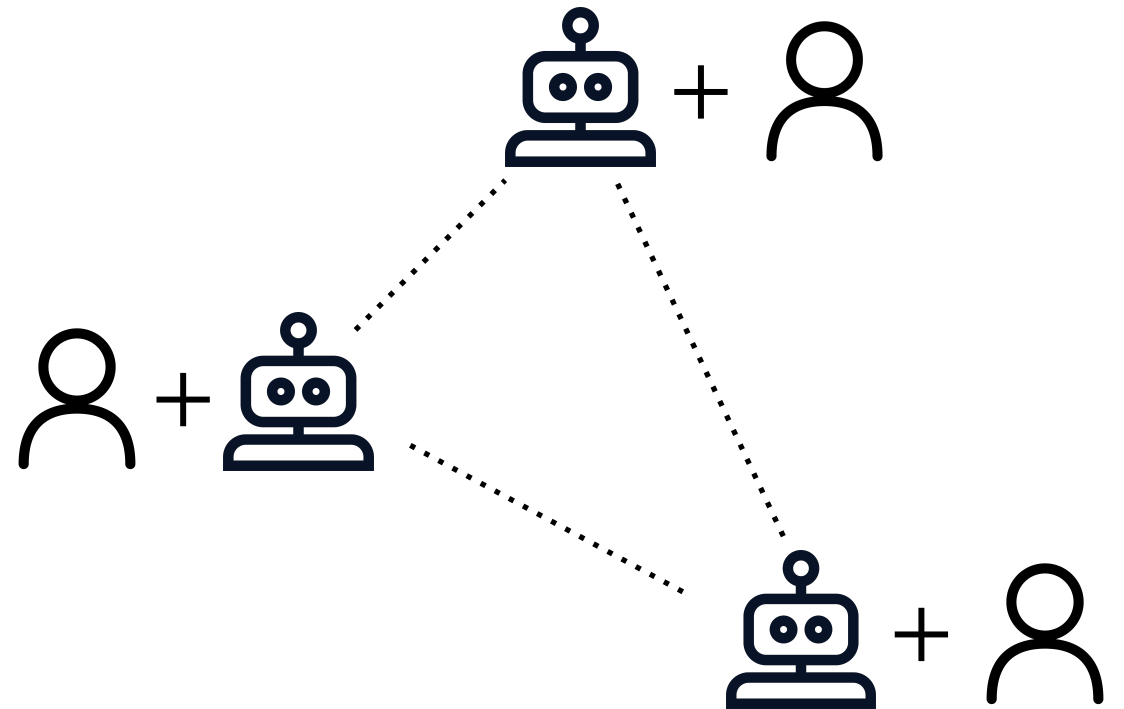
The Human+AI unit



Contracting, Delegation

Level 3:

A society/market of
human+AI units



Games, Markets

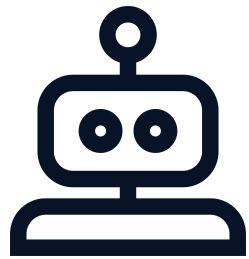
Level 1: The AI Itself

How do we conceptualize the AI?

- As an (economic) **agent**

(Immorlica, Lucier, and Slivkins 24, Raman, Lundy, Amouyal, Levine, Leyton-Brown, and Tennenholtz 24)

AI system



- As a **tool**

(Agrawal, Gans, and Goldfarb 19, Hewitt, Ashokkumar, Ghezae, and Willer 24)

- As an **aggregation**/reflection of human behavior

(Aher, Arriaga, and Kalai 23, Horton, Filippas, and Manning 26, Horton and Manning 26)

Need approaches for **measurement** and **evaluation**

(Fish, Shepherd, Li, Shorrer, Gonczarowski 26, Gao, Han, and Liang 26)

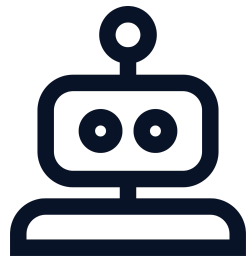
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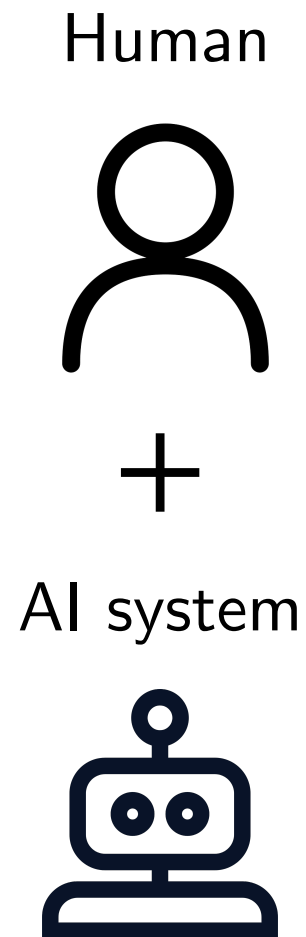
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A measure of their pretrained knowledge
(with Wayne Gao and Sukjin Han)

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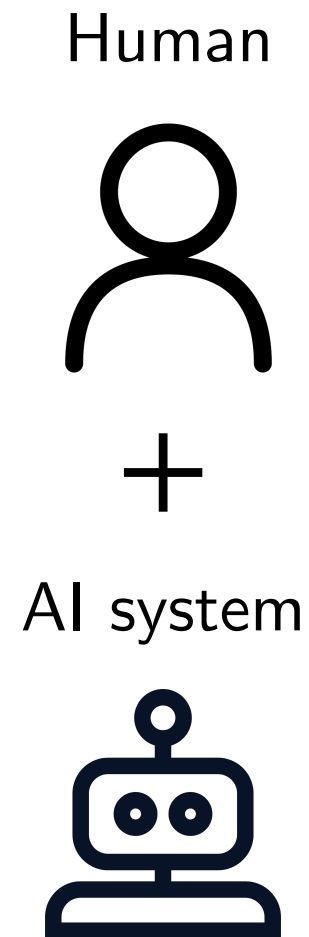
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Level 2: Human + AI



- What is the human-AI **production function**?
(Kleinberg et al. 18, Agarwal et al. 23, Martin, Kovach, and Tserenjigmid 26, Almog 26, Goldsmith-Pinkham, Tan, and Zentefis 26, Dell'Acqua et al 26)
- How do we **design** effective human-AI collaboration?
 - **Over/under-reliance** on AI
(Agarwal, Moehring, and Wolitzky 26, Hoong and Dreyfuss 26, Amin et al. 26)
 - The **alignment** problem
(Chen, Ghersengorin, and Petersen 24, Ge, Halpern, Micha, Procaccia, Shapira, Vorobeychik, Wu 24; Gölz, Haghtalab, and Yang 25, Collina, Goel, Roth, Ryu, and Shi 25, Fudenberg and Liang 26, Dworczak and Smolin 26)
- Dynamic **effects** and **externalities**: How does AI use
 - change human skills?
(Bondi and Johnson 26, Fudenberg and Levine 26, Aouad, Lykouris, and Zhong 26, Acemoglu, Dong, and Ozdaglar 26)
 - the incentives to create?
(Garicano and Rayo 26, Li, Li, and Cullen 26, Liang and Lu 26)

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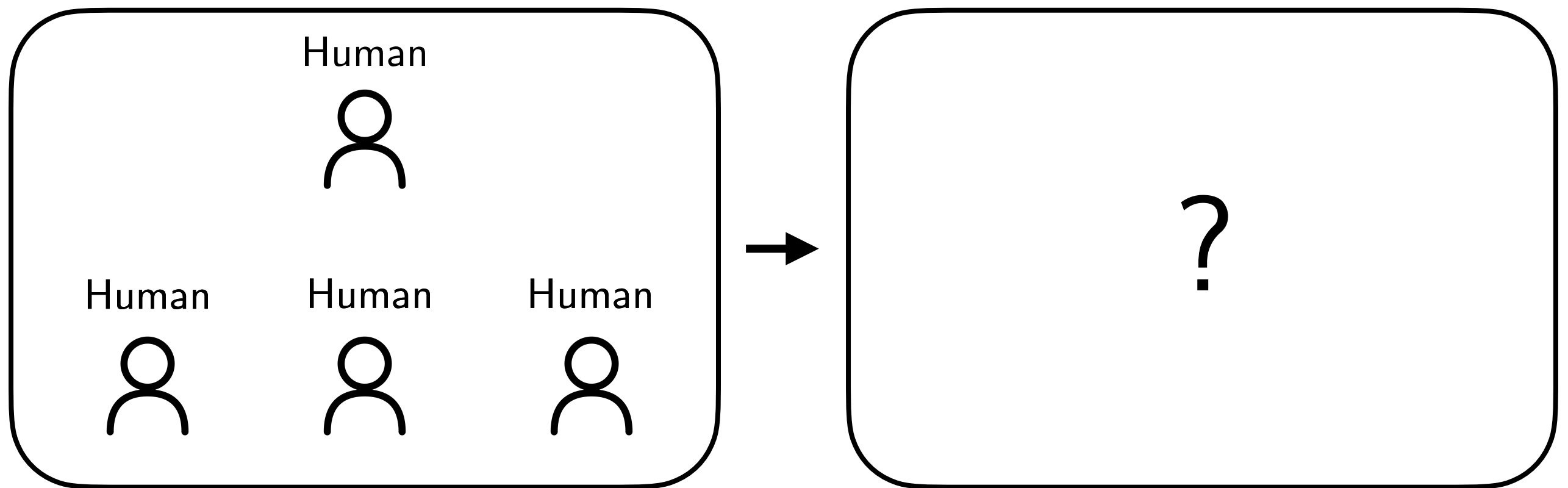
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Paper 2: Creative Ownership in the Age of AI
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Level 3: Many Human-AI Systems

What happens when AI-mediated agents interact with one another?

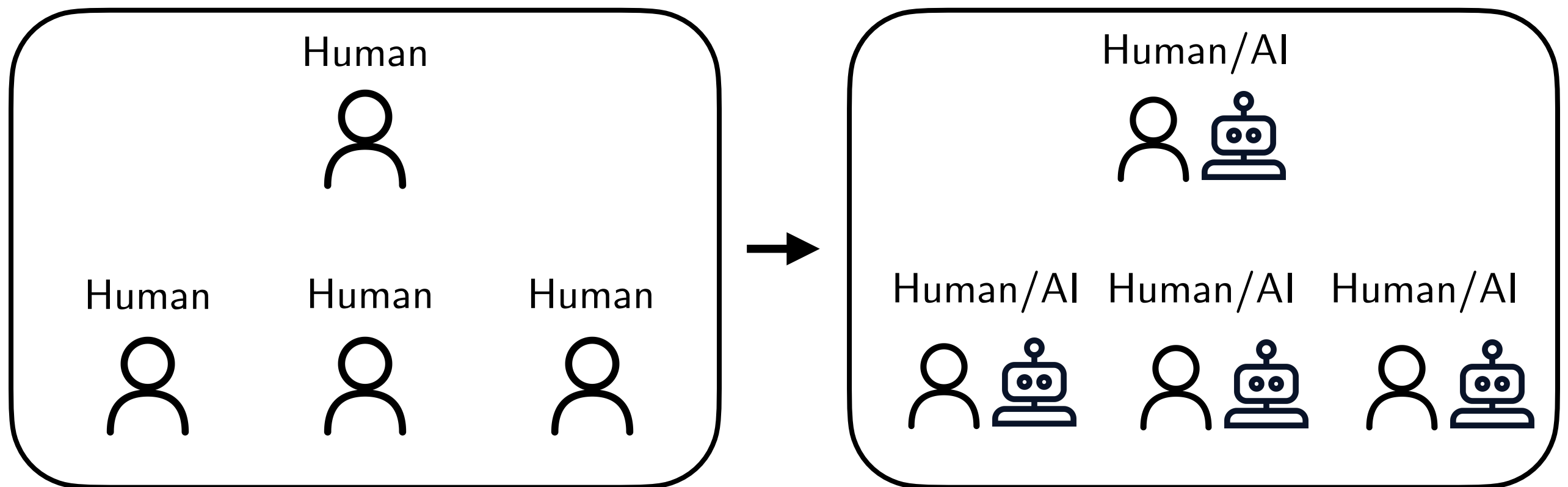
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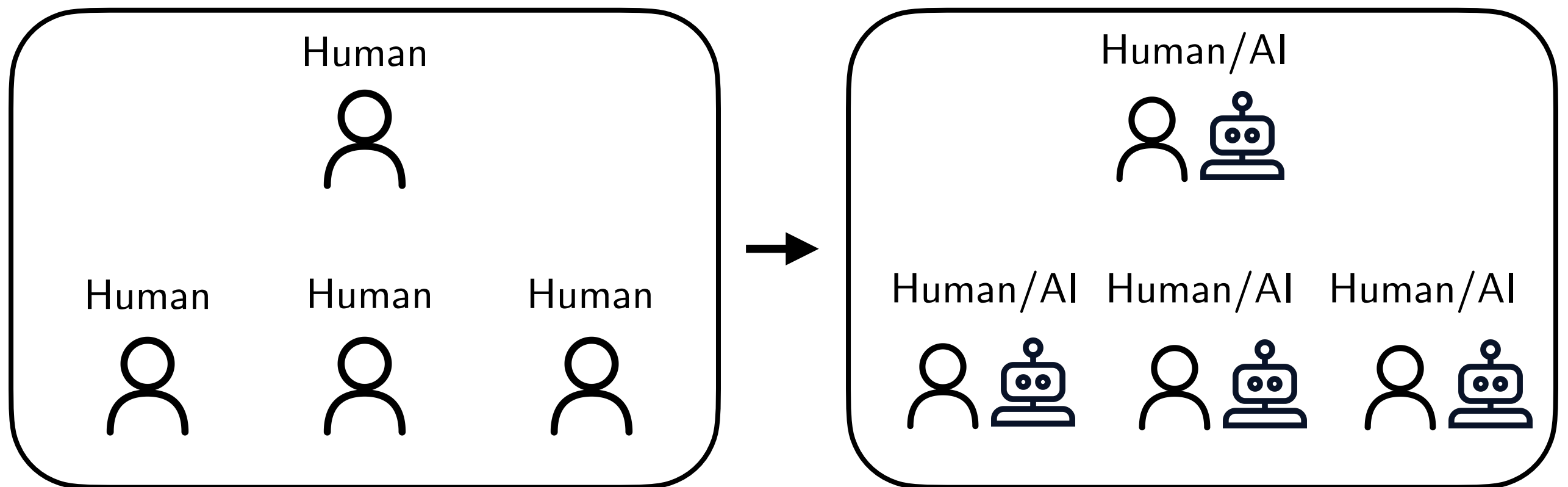


Level 3: Many Human-AI Systems

What happens when AI-mediated age

Paper 3: AI and the Production of Research

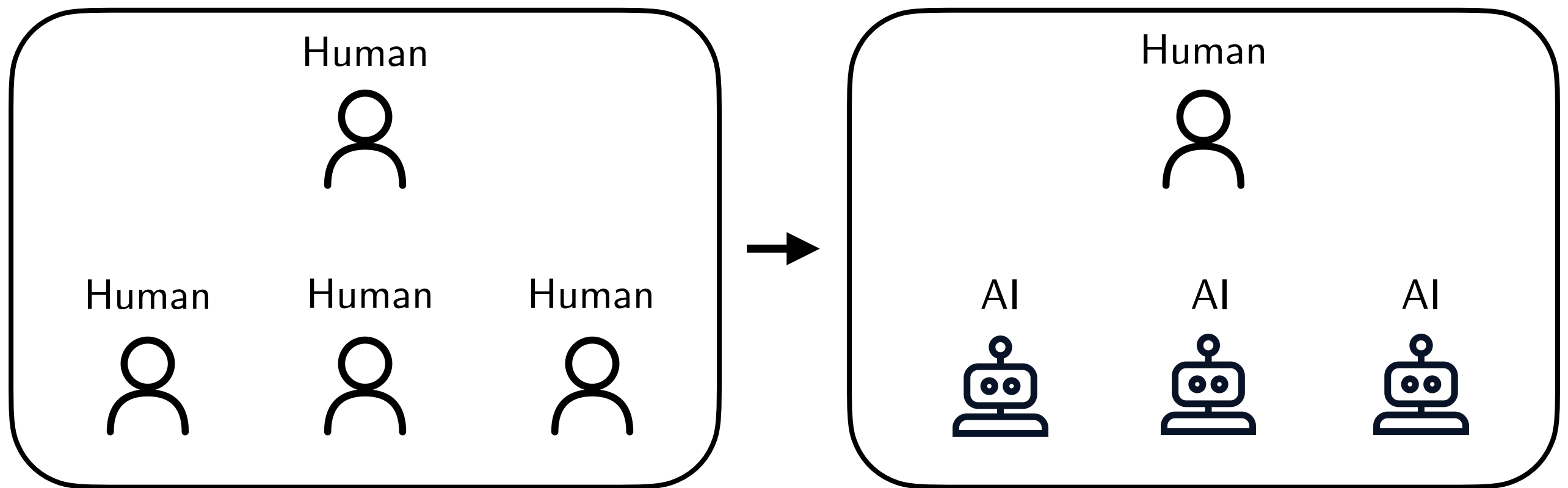
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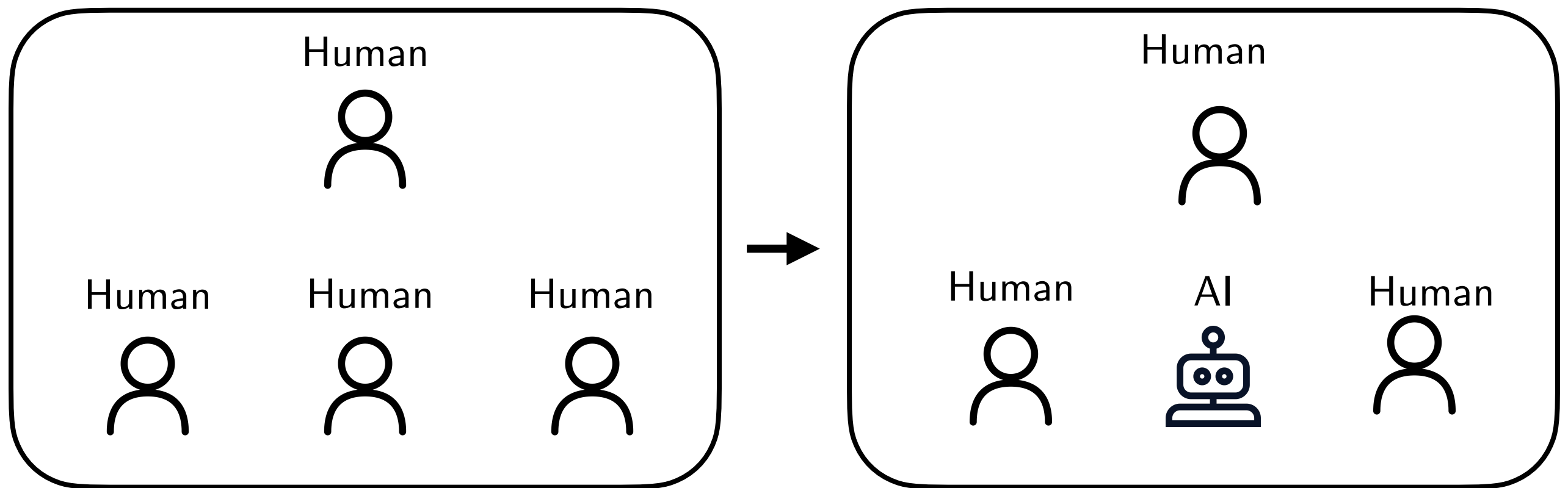
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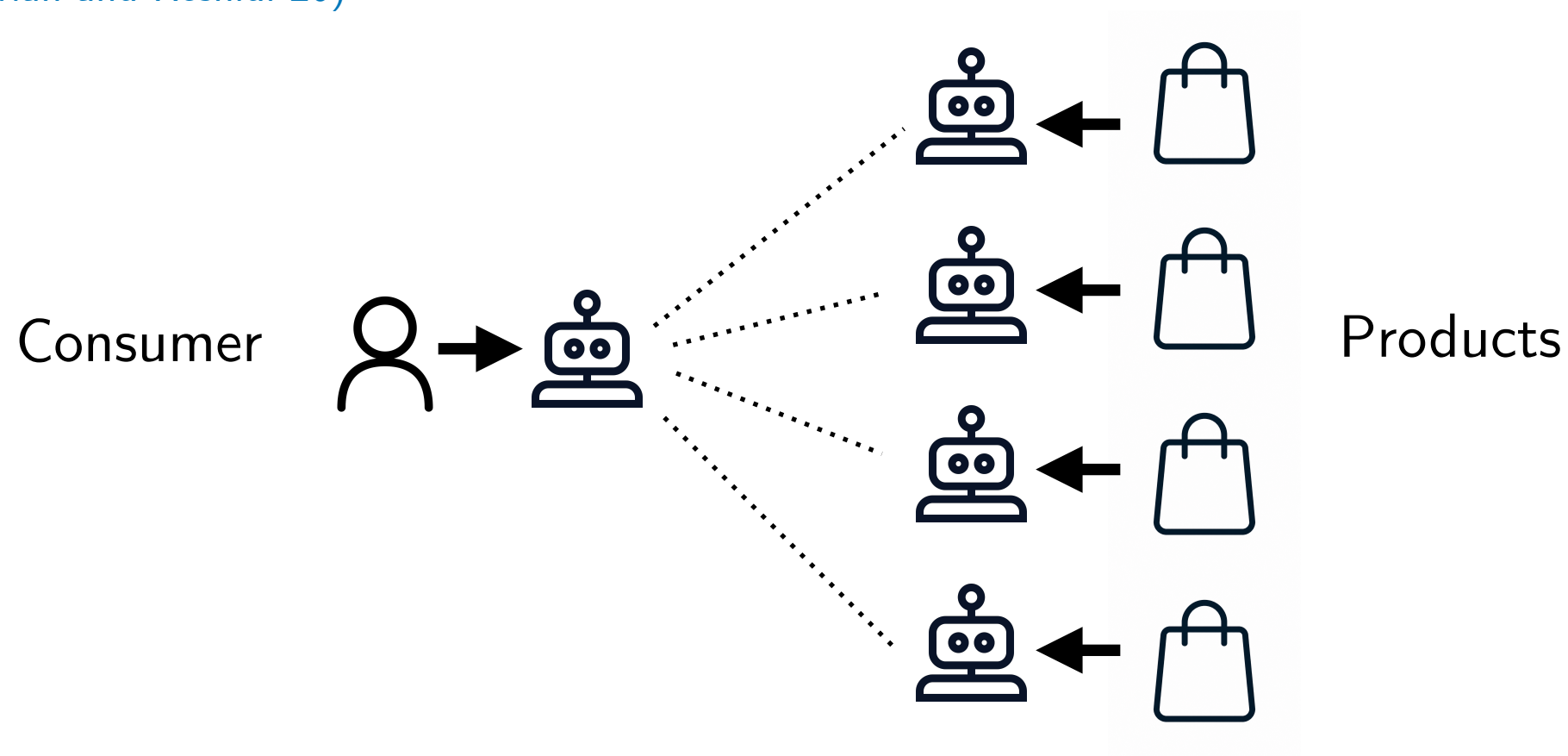
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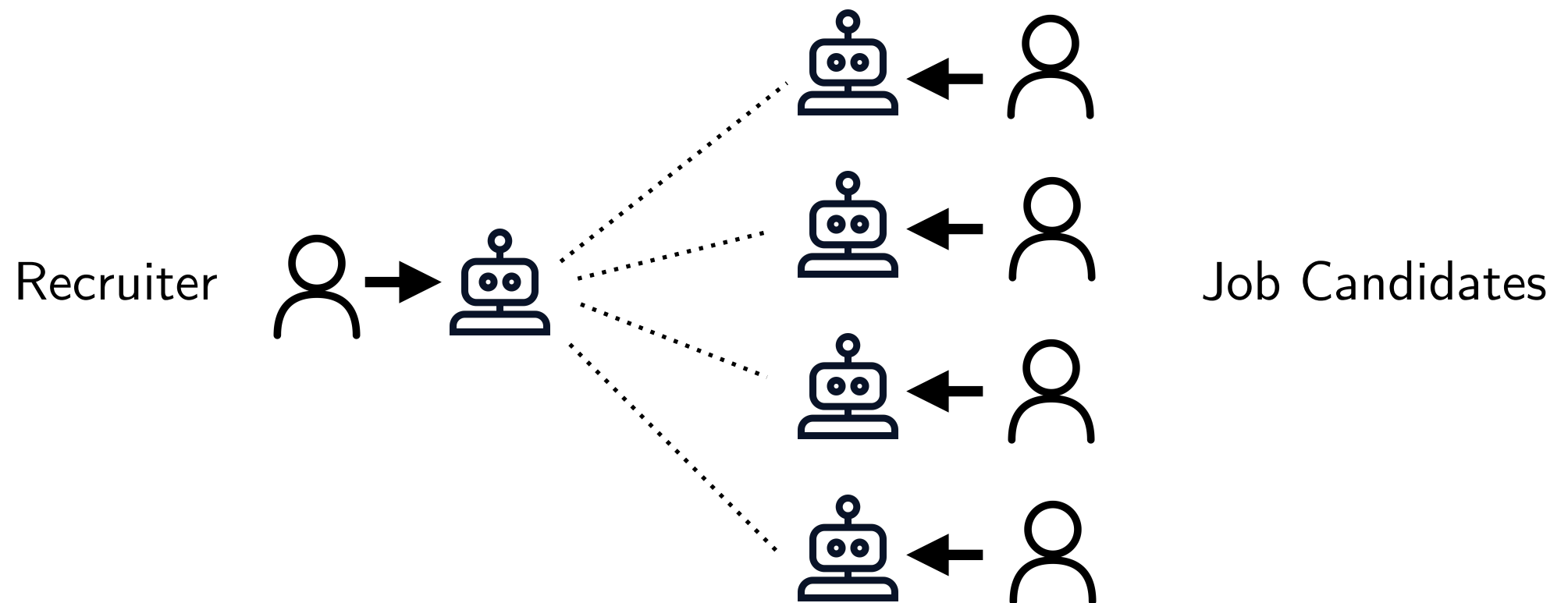
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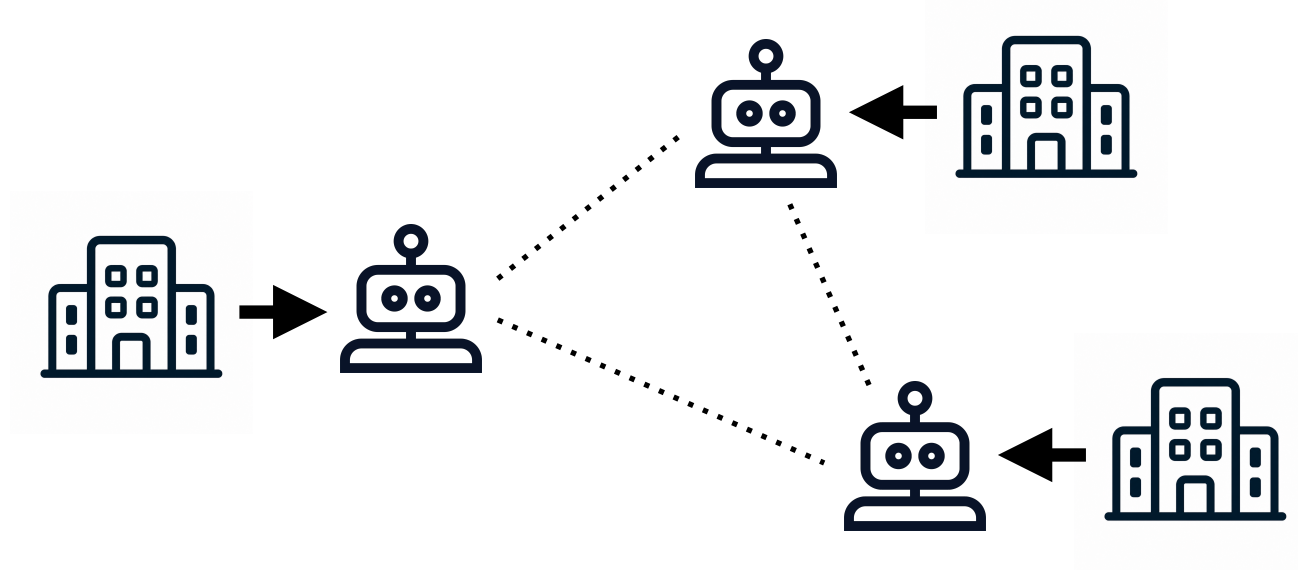
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(Kleinberg and Raghavan 21, Henkel and Jabarian 25, Liang 25, Lucier et al. 26, Ellis and Huang 26, Jabarian and Reshidi 26)
- What happens when organizations use algorithms to set prices/bid/compete?
(Calvano et al. 20, Banchio and Skrzypacz 22, Banchio Mantegazza 26, Fish, Gonczarowski, Shorrer 26)



Part II

Papers

Papers

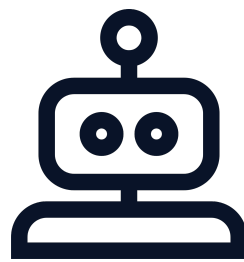
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2. Creative Ownership in the Age of AI (with Jay Lu)
3. AI and the Production of Research (with Tylor Wu)
4. Artificial Intelligence Clones

How Well Do LLMs Predict Human Behavior? A Measure of their Pretrained Knowledge

Wayne Gao
(Penn)

Sukjin Han
(Bristol)

Annie Liang
(Northwestern)



Introduction

A striking feature of LLMs:

- mimic human behaviors across a wide range of domains
- because of extensive **pretraining** on human-generated text
- e.g., LLMs reproduce human behavioral biases (anchoring, framing, loss aversion) (Jones & Steinhardt 22, Horton 23)

Thus increasingly used to **predict** human behaviors, e.g.,

- predicting outcomes of economic experiments (Aher et al 23, Hewitt et al 24, Shapira et al 25, Park et al 24, Horton and Manning 26)
- forecasting macroeconomic variables (Bybee 23, Faria-e-Castro and Leibovici 24, Carriero et al 25)

Example

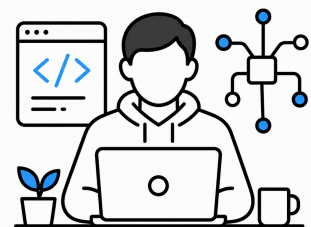
Problem: Which action do row players of this game choose most often?

	a_1	a_2	a_3
a_1	25, 25	30, 40	100, 31
a_2	40, 30	45, 45	65, 0
a_3	31, 100	0, 65	40, 40



ECONOMIST

Choose your favorite game theory model and solution concept, and use data of play in games to estimate the free parameters



MACHINE
LEARNER

Use data of play in games to train a machine learning algorithm (e.g., a neural net) that maps payoff matrices into actions

Example

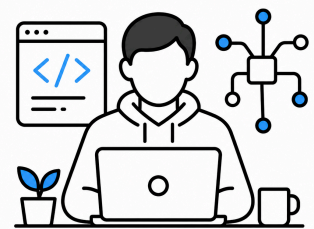
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User of LLMs

	a_1	a_2	a_3
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Which action do you think would be most frequently chosen by subjects playing in the role of the row player of this 3x3 normal form game?



Thought for 39s >

I would predict a_2 would be chosen most frequently.

Example

NO DOMAIN-SPECIFIC DATA REQUIRED!

...but are these predictions any good?



User of LLMs

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Proposed Measure

- We need a way to evaluate how useful the LLM is
- One option is simply “predictive accuracy”
- Two issues with this:
 - Predictive accuracy can be hard to interpret (is 2.4 RMSE small?)
 - The LLM exists in the context of alternative approaches to prediction, so its **relative** value is often what is relevant
- Key tradeoff: with enough domain-specific data, flexible algorithms will outperform the LLM, but that data is costly

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- Key tradeoff: with enough domain-specific data, flexible algorithms will outperform the LLM, but that data is costly
- Our proposed measure: the LLM’s **equivalent sample size**, defined as how much data is needed for the algorithm to beat the LLM
- Can be interpreted as a proxy for “cost saved”

Setting

Prediction problem:

- covariate vector $X \in \mathcal{X}$
- outcome of interest $Y \in \mathcal{Y}$ (could be a distribution)

A **prediction rule** is a mapping $f: \mathcal{X} \rightarrow \mathcal{Y}$

Measure prediction accuracy via a loss function $\ell: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$

The **risk** (expected loss) of prediction rule f is

$$e(f) := \mathbb{E}_P[\ell(f(X), Y)]$$

where P is the true (unknown) distribution of (X, Y)

Two Prediction Approaches

LLM: Fix an LLM and prompting protocol that

- describes the prediction task
- gives the LLM X in natural language and asks it to predict Y

Denote the resulting prediction rule by f_{LLM} and its risk by $e_{LLM} \equiv e(f_{LLM})$

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Domain-Specific Learning: An algorithm (or economic model)

$a : \mathcal{D} \rightarrow \mathcal{F}$ maps data to a prespecified set of prediction rules

- e.g., use the data to estimate the parameters of an economic model or ML algorithm by minimizing empirical risk

Define $f_N \equiv a(D_N)$ to be the prediction rule we get from applying a to a (random) dataset of size N . Define $e_N^a \equiv \mathbb{E}[e(f_N)]$ to be its expected risk.

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No dependence on N

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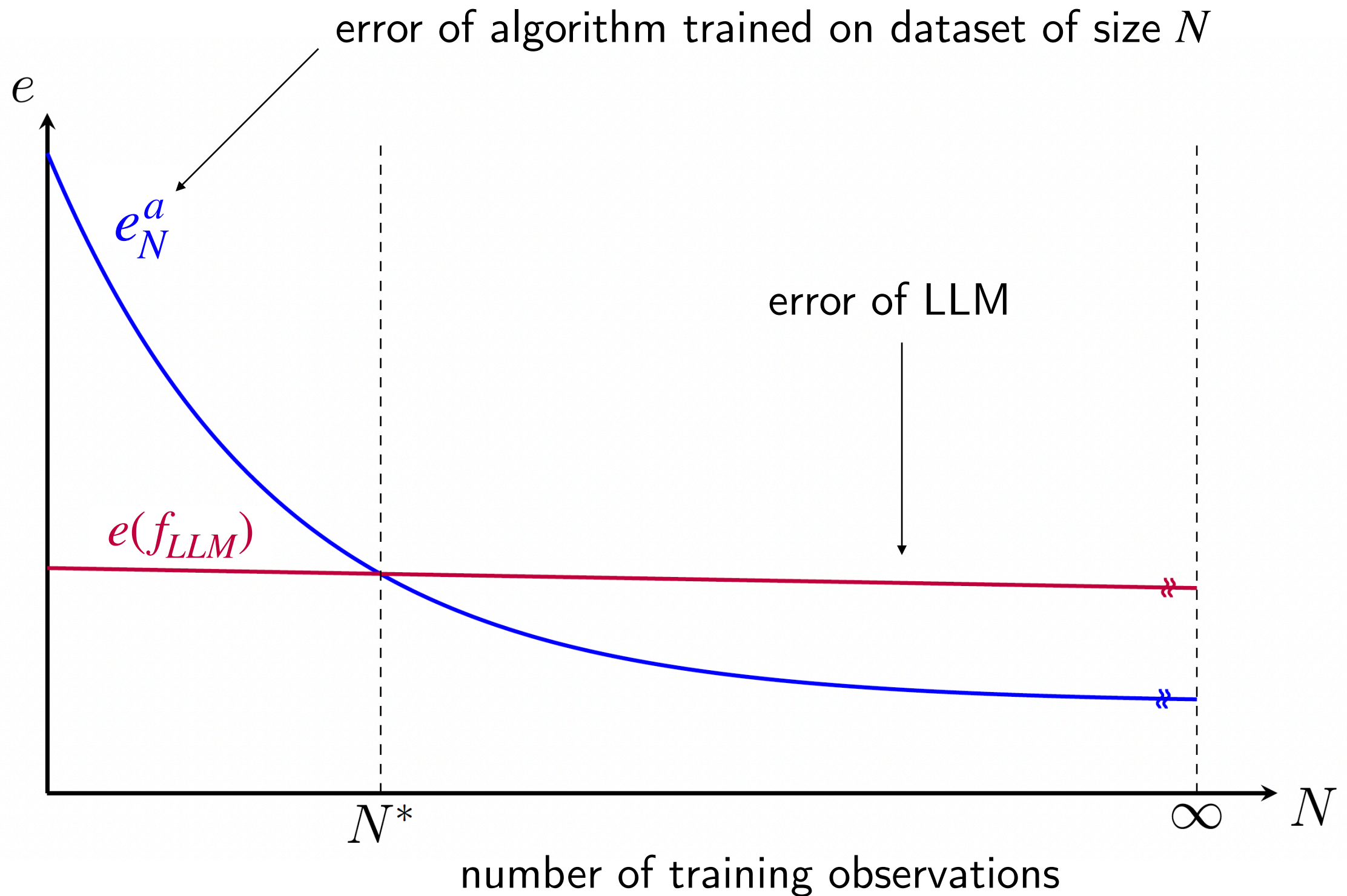
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Domain-Specific Error Curve



α -Equivalent Sample Size (ESS)

Definition: The LLM's α -equivalent sample size (ESS) is

$$N^* = \min\{N : e_N^\alpha \leq e_{LLM}\}$$

- smallest training size such that the algorithm/model outperforms the pretrained LLM
- interpret as the minimum amount of domain-specific data the analyst would need to beat the LLM
- large $N^* \Rightarrow$ the LLM provides substantial predictive value in this domain; small $N^* \Rightarrow$ it does not

Econometric Contributions

The paper provides a method for estimating and doing inference on N^* given a sample of (x, y) observations (skip details today)

- Estimate e_{LLM} by computing empirical risk on that sample
- Estimate e_N^a using block-out cross-validation:
 - Split the sample into blocks of size N , train on one block and test on the remaining $N - 1$, average across blocks
 - Characterize asymptotic distribution of this blockout cross-validated estimator

Plugin estimator for N^* : first N at which our estimate of e_N^a beats our estimate of e_{LLM} .

Application

Classification problem: given a 3x3 payoff matrix, predict the action most likely to be chosen by the row player.

	a_1	a_2	a_3	
a_1	25, 25	30, 40	100, 31	$\longrightarrow$ $\{a_1, a_2, a_3\}$ Y
a_2	40, 30	45, 45	65, 0	
a_3	31, 100	0, 65	40, 40	

X

Our analysis: evaluate the equivalent sample size of GPT 5.4 relative to:

- A popular behavioral game theory model (Poisson Cognitive Hierarchy)
- Random Forest

Data

We consider two datasets:

- **Dataset A:** play in 86 games, aggregated across six experimental game theory papers (Wright and Leyton-Brown 14)

Data

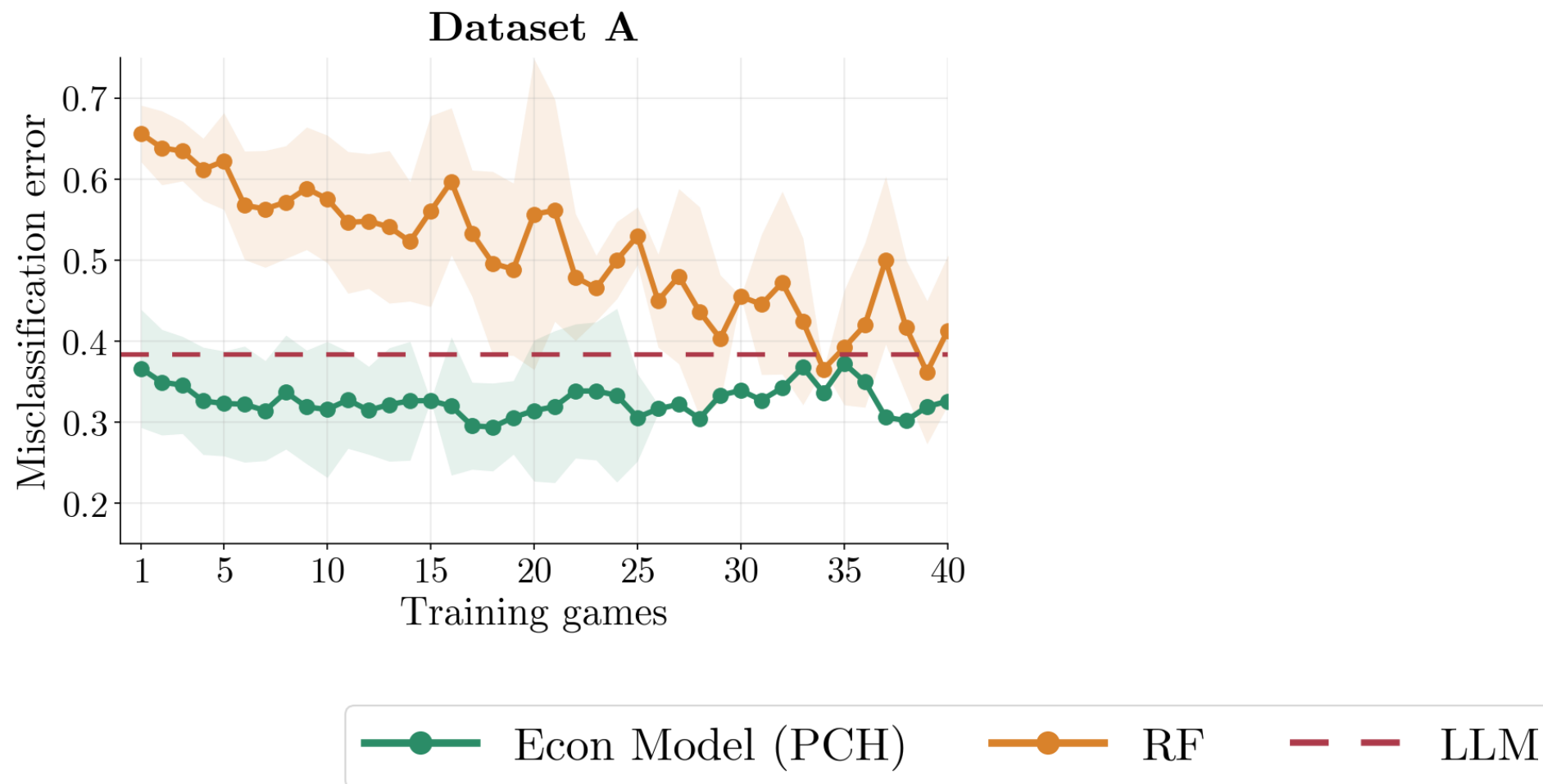
We consider two datasets:

- **Dataset A:** play in 86 games, aggregated across six experimental game theory papers (Wright and Leyton-Brown 14)
- **Dataset B:** play of 200 games, algorithmically generated to be strategically rich and different from those in Dataset A (Fudenberg and Liang 19)

Example:

	a_1	a_2	a_3	
a_1	90, 90	30, 80	45, 30	
a_2	80, 30	55, 55	37, 5	← Level 1
a_3	30, 45	5, 37	70, 70	

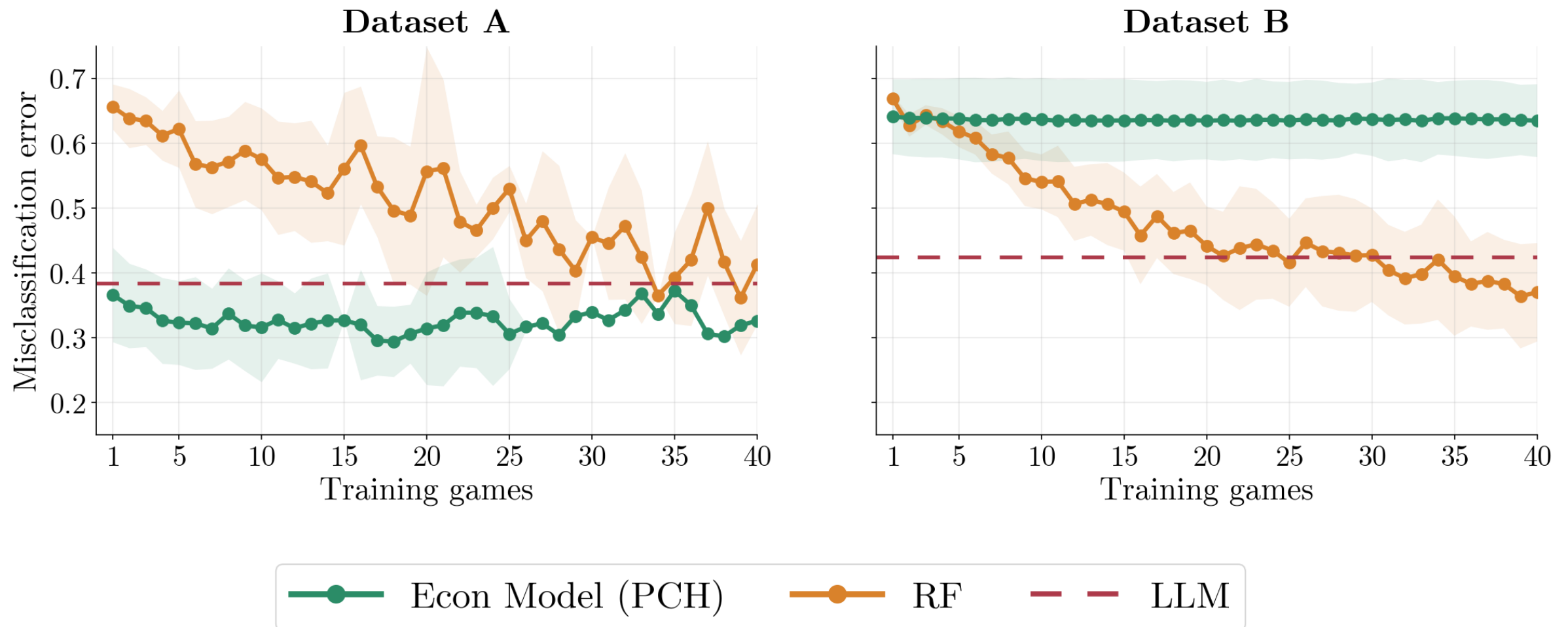
Results



On the lab games designed by experimental game theorists:

- ESS very low with respect to the economic model (Poisson Cognitive Hierarchy Model): plugin estimate $\hat{N}^* = 1$, confidence interval of $[1, \infty)$

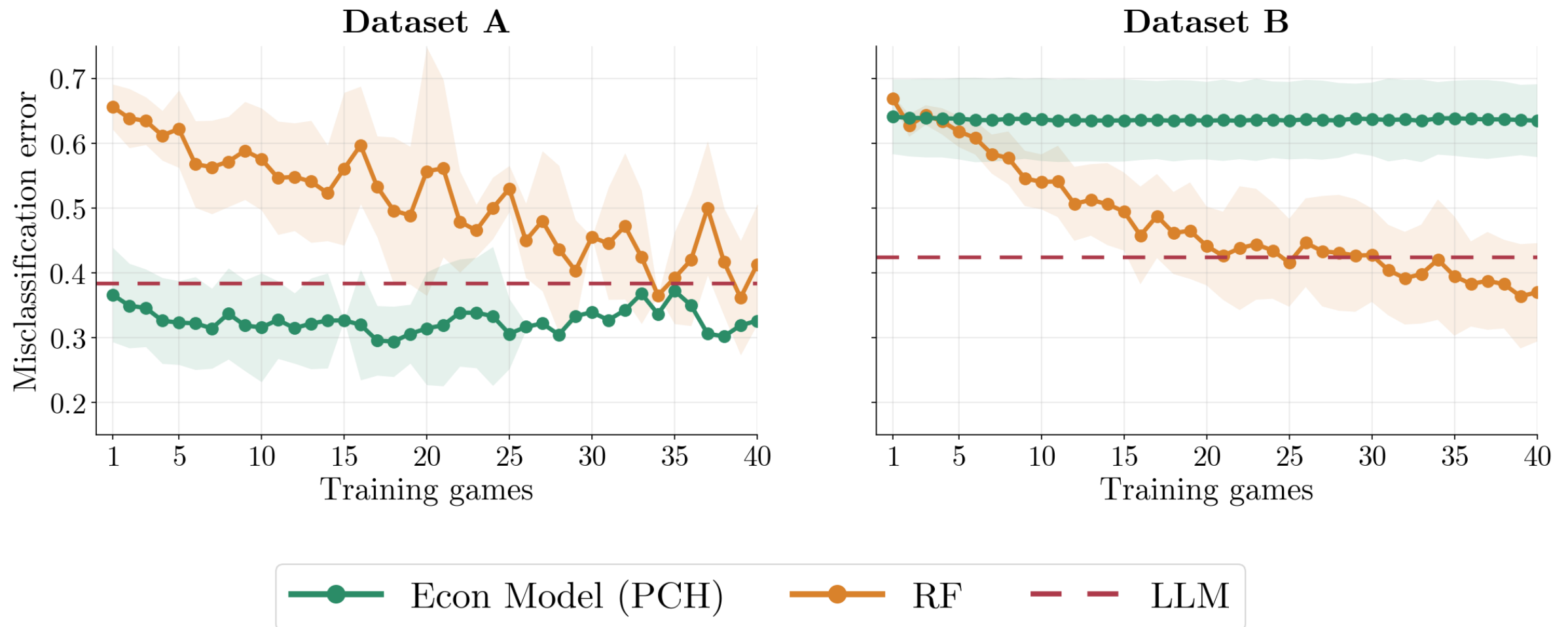
Results



In the new games:

- The LLM does nearly as well as it did on the lab games, but the economic model does much worse, and stays worse
- Eventually the ML algorithm surpasses the LLM (plugin estimate $\hat{N}^* = 25$, confidence interval of $[16, \infty)$)

Results

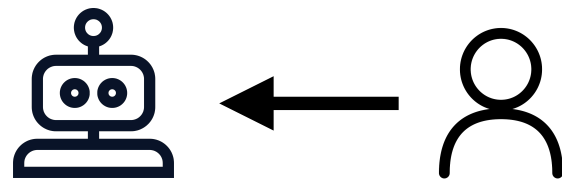


Takeaway: LLMs can be valuable when our theories fail

Creative Ownership in the Age of AI

Annie Liang
(Northwestern)

Jay Lu
(UCLA)



Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

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- Novels

The AI-generated Books Trend is Getting Worse

It has gotten so bad that Amazon's new policy limits self-published Kindle ebooks to three a day.



Sascha Brodsky, Contributor
January 24, 2024

⌚ 4 Min Read

AI BUSINESS

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Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
- Cartoons

Generating New Yorker-Style Cartoons with AI

Harnessing the power of OpenAI's ChatGPT to Transform Cartoon ideas into Reality



Georgiy Ermakov

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5 min read · Oct 25, 2023

Introduction

Generative AI systems—trained on large corpora of human-authored works—are increasingly used to produce new creative content.

- Novels
- Cartoons
- Even AI “actors”

Tilly Norwood: how scared should we be of the viral AI ‘actor’?

A bunch of code is being pushed as the next Scarlett Johansson, a creation that is already causing pushback from real human actors

Tilly Norwood



A 2025 AI-generated image of Tilly Norwood, taken from the character's [Instagram](#) account

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- Cartoons
- Even AI “actors”

These creations can closely imitate style without copying content, a capability now central to ongoing litigation.

- Andersen v. Stability AI
 - Disney Enterprises v. Midjourney
 - Universal Music Group Recordings v. Suno
- + many others

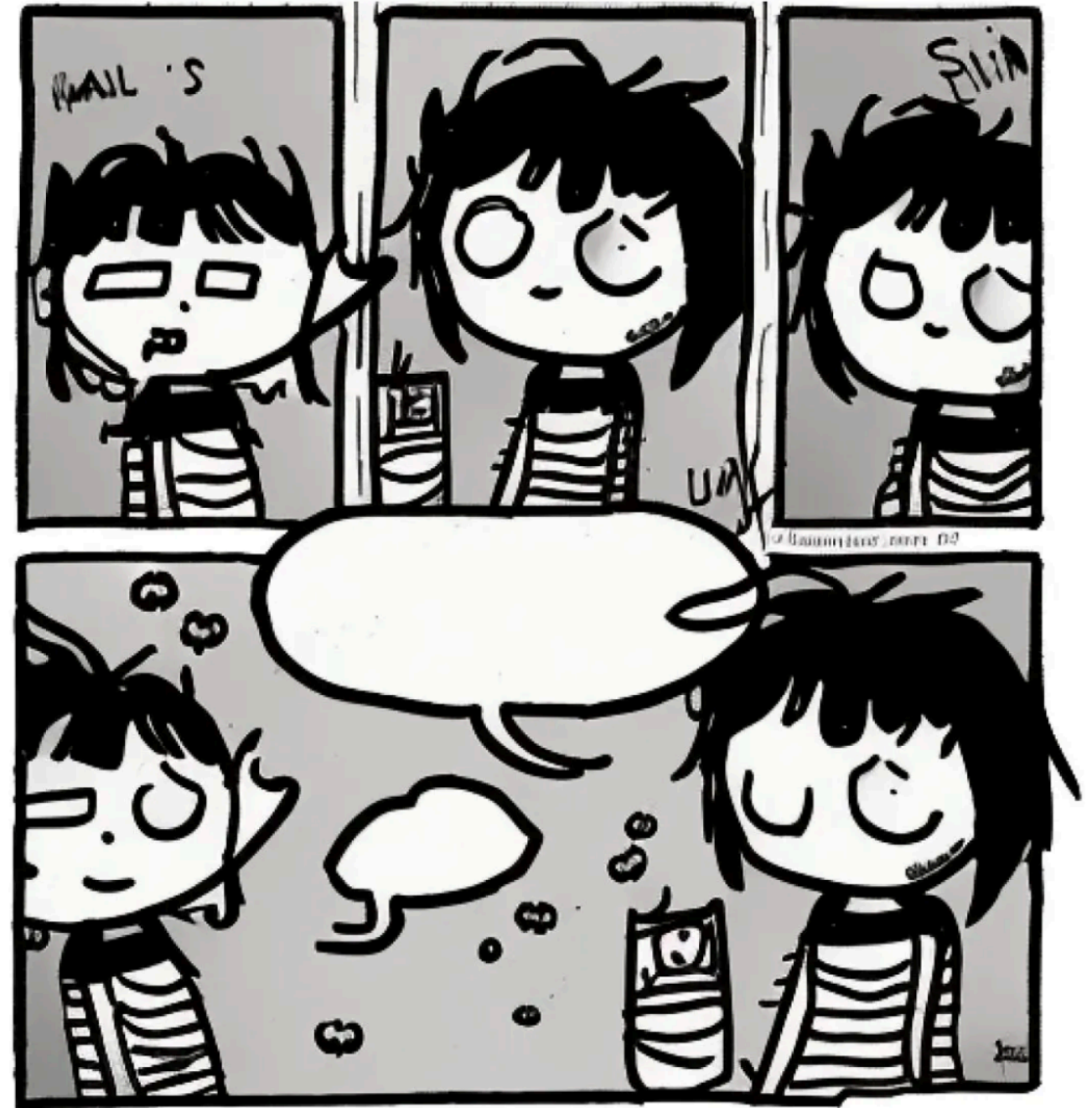
Introduction

A recent example of my work



Sarah Andersen, *NYTimes*, 2022

An A.I. generated image that I created when I used my name as a prompt



It's not perfect — but it has captured the signature elements of my drawing style. The black bangs, striped shirt and wide eyes are immediately recognizable. As A.I. gets increasingly sophisticated, I am growing more concerned about what's possible.

“Substantial Similarity”

Generated outputs may violate copyright law if they are “substantially similar” to an existing work.

- Historically interpreted in a narrow sense to preserve creative competition and follow-on innovation.
- But did not account for the emergence of AI systems that could easily, flexibly, quickly mimic style

This Paper

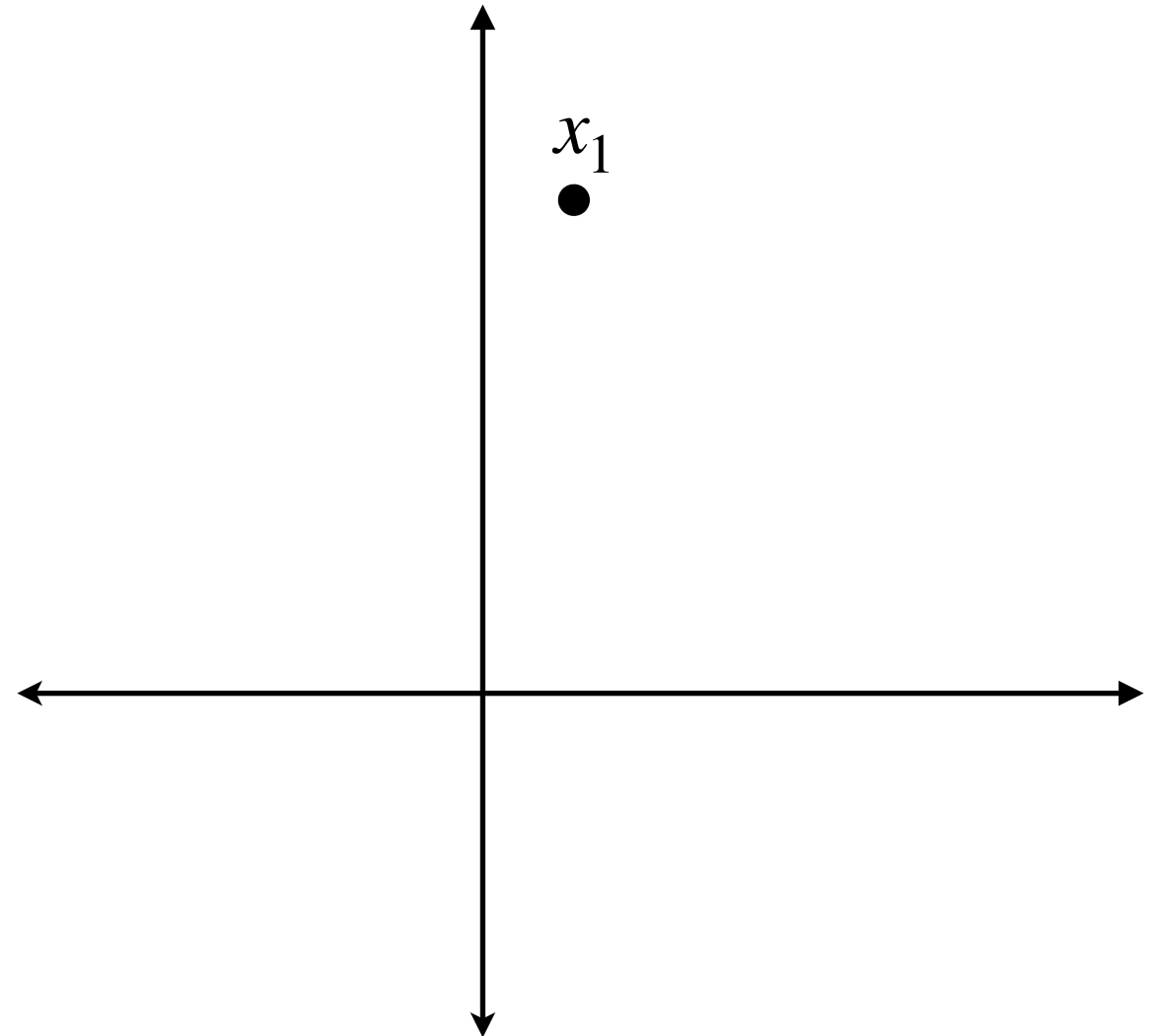
Existing notions of copyright and intellectual property **may not be appropriate for a post-generative AI world.**

We propose a new criterion for evaluating when an output is derivative of an existing work:

Would the output have been generable without that existing work in its training corpus?

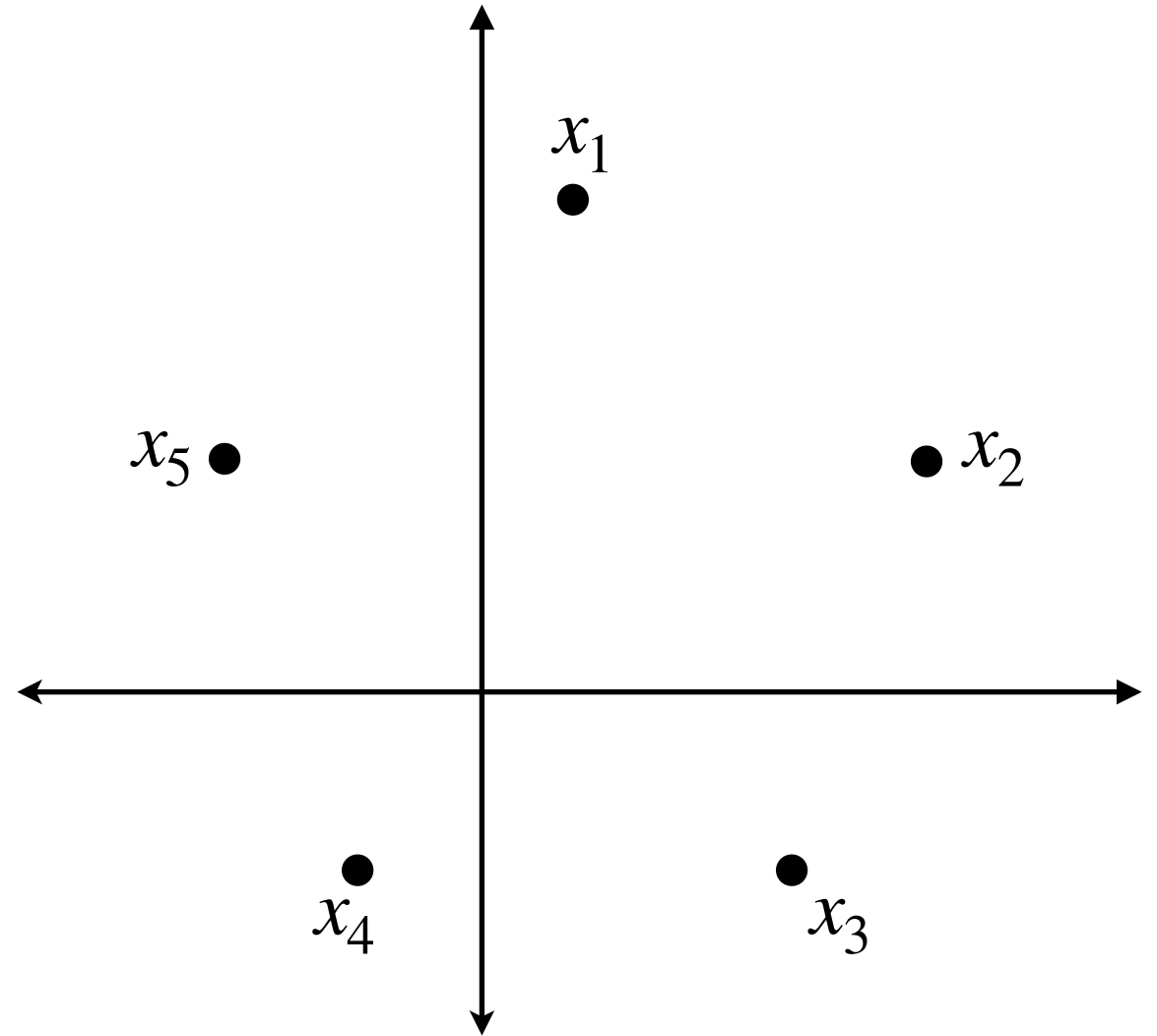
Model

- A *creation* is a point $x \in \mathbb{R}^d$



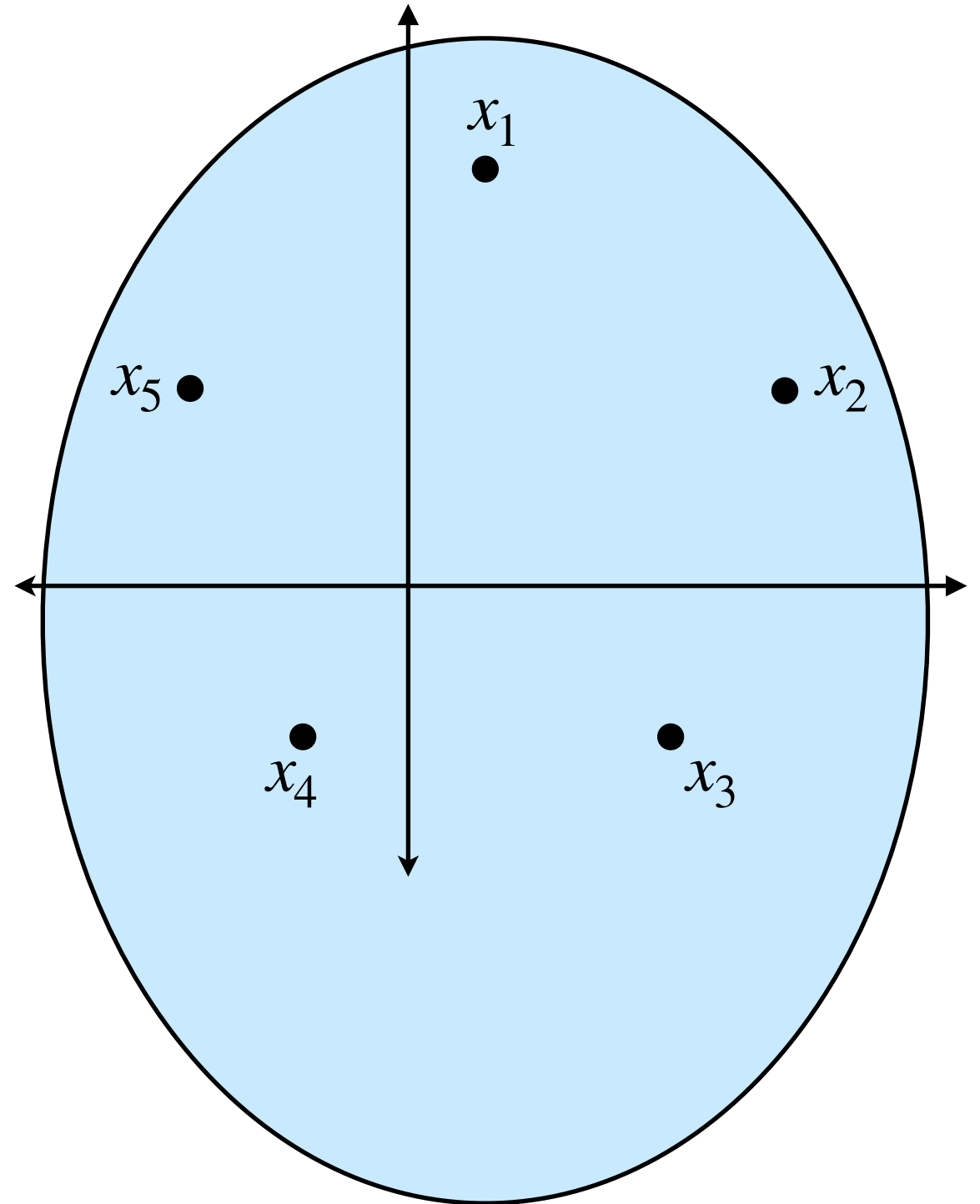
Model

- A *creation* is a point $x \in \mathbb{R}^d$
- A *corpus* is a set of creations



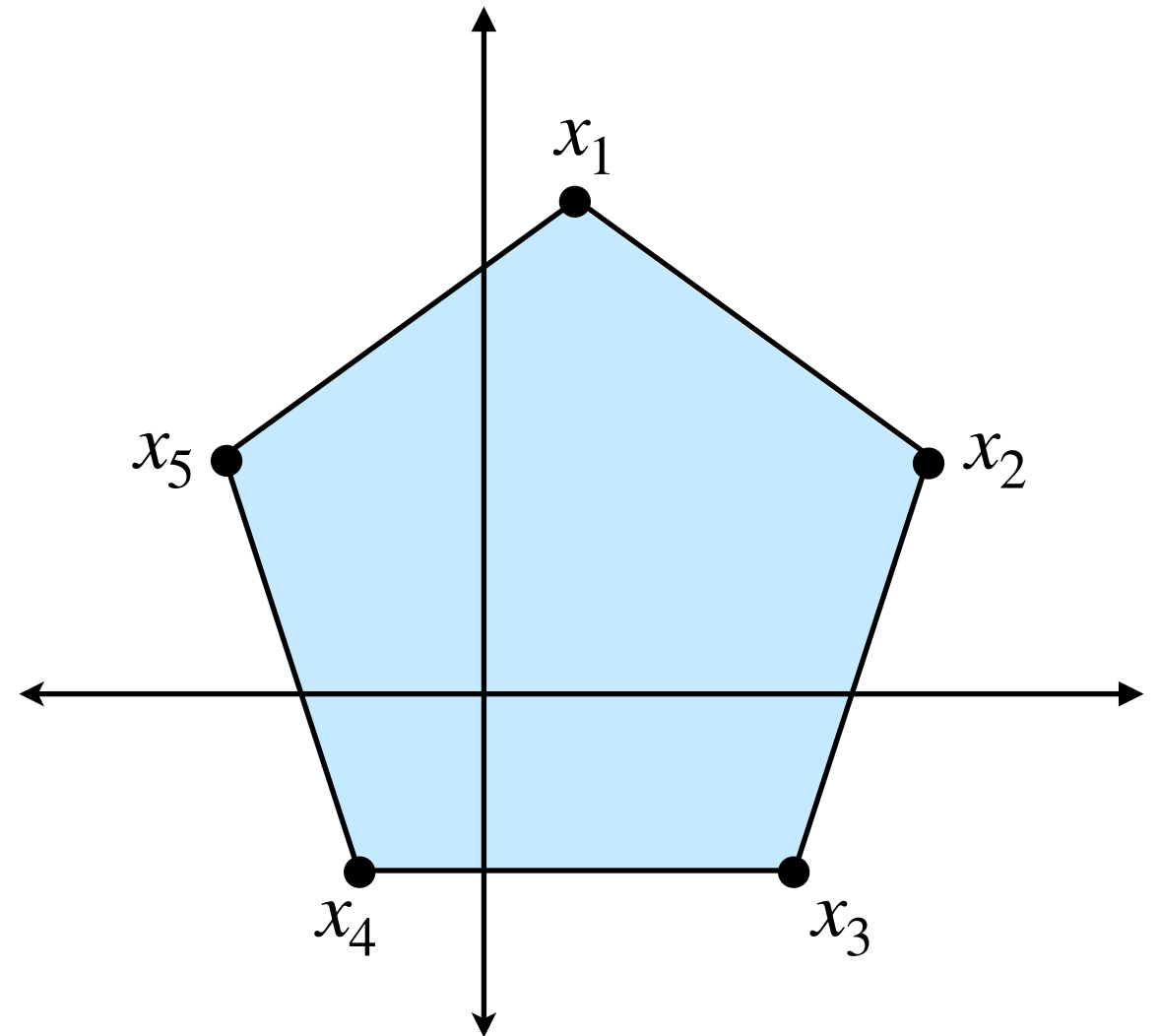
Model

- A *creation* is a point $x \in \mathbb{R}^d$
- A *corpus* is a set of creations
- A *generator* is a technology for producing new creations from an existing corpus, satisfying:
 1. $C \subseteq g(C)$
 2. $C \subseteq C'$ implies $g(C) \subseteq g(C')$
 3. $g(g(C)) = g(C)$



Model

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 1. $C \subseteq g(C)$
 2. $C \subseteq C'$ implies $g(C) \subseteq g(C')$
 3. $g(g(C)) = g(C)$
- The generator is *convex-valued* if every $g(C)$ is convex (minimal case is the convex-hull generator)



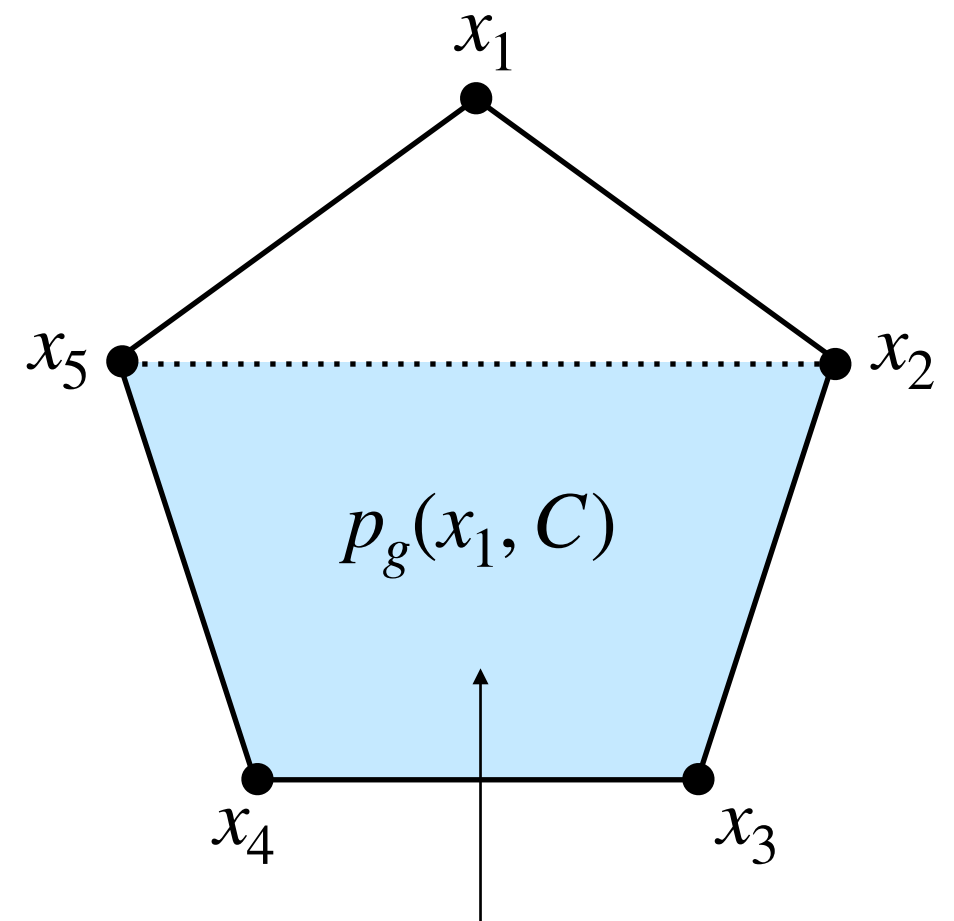
Permissible Generation

Definition: Fix any generator g , corpus C , and work $x \in C$. The x -**permissible set** is

$$p_g(x, C) := g(C \setminus \{x\})$$

i.e., all creations that could have been generated without x

Convex hull generator:



x_1 -permissible set

(generable creations that do not require x_1)

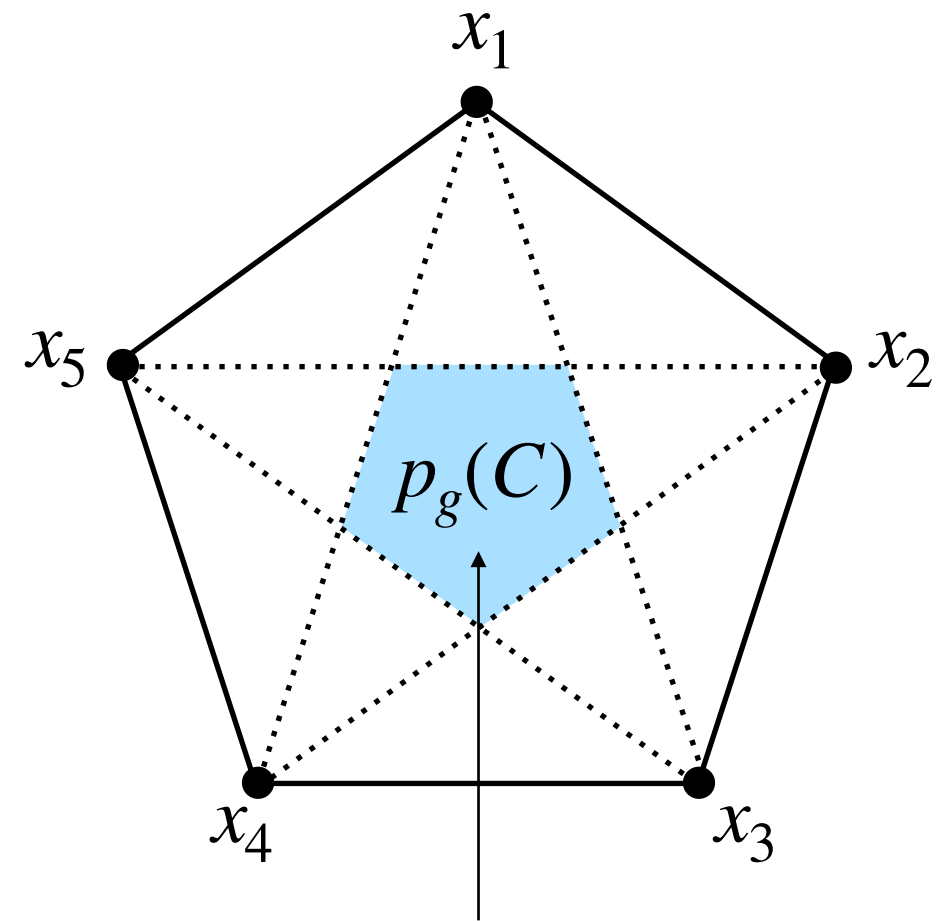
Permissible Generation

Definition: Fix any generator g and corpus C . The **permissible set** is

$$p_g(C) := \bigcap_{x \in C} p_g(x, C)$$

i.e., all creations that do not critically depend on any $x \in C$

Convex hull generator:



permissible set

(generable creations that do not critically rely on any $x \in C$)

Plan

In the paper: structural properties of the permissible set

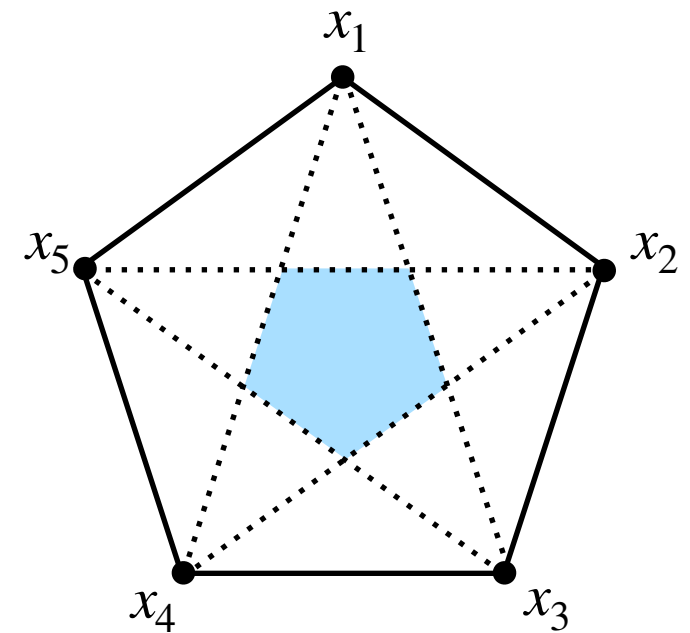
Today focus on what the permissible set looks like when the corpus is large
—will individual creations retain their importance?

Permissible Ratio

Let $Vol(A)$ denote the Lebesgue measure (or *volume*) of the set $A \in \mathcal{C}$.

Definition: The **permissible ratio** is:

$$r_g(C) := \frac{Vol(p_g(C))}{Vol(g(C))}$$

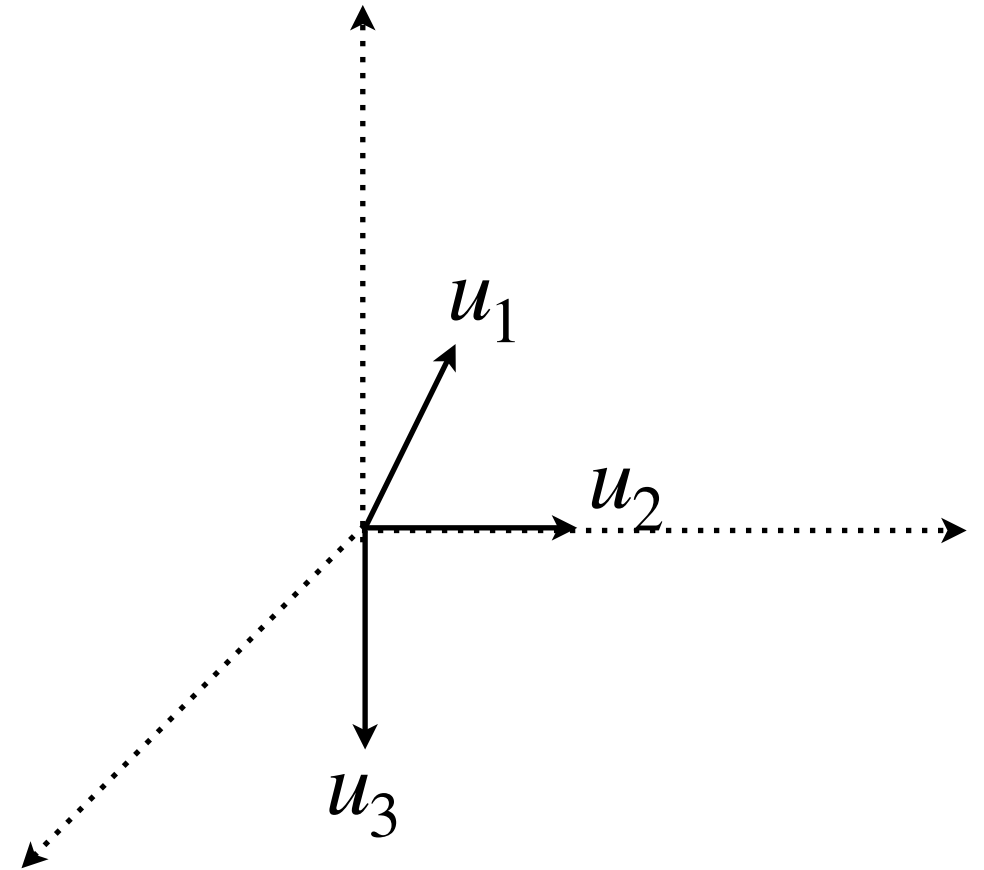


i.e., the proportion (by volume) of generable creations that are permissible.

Corpus Formation

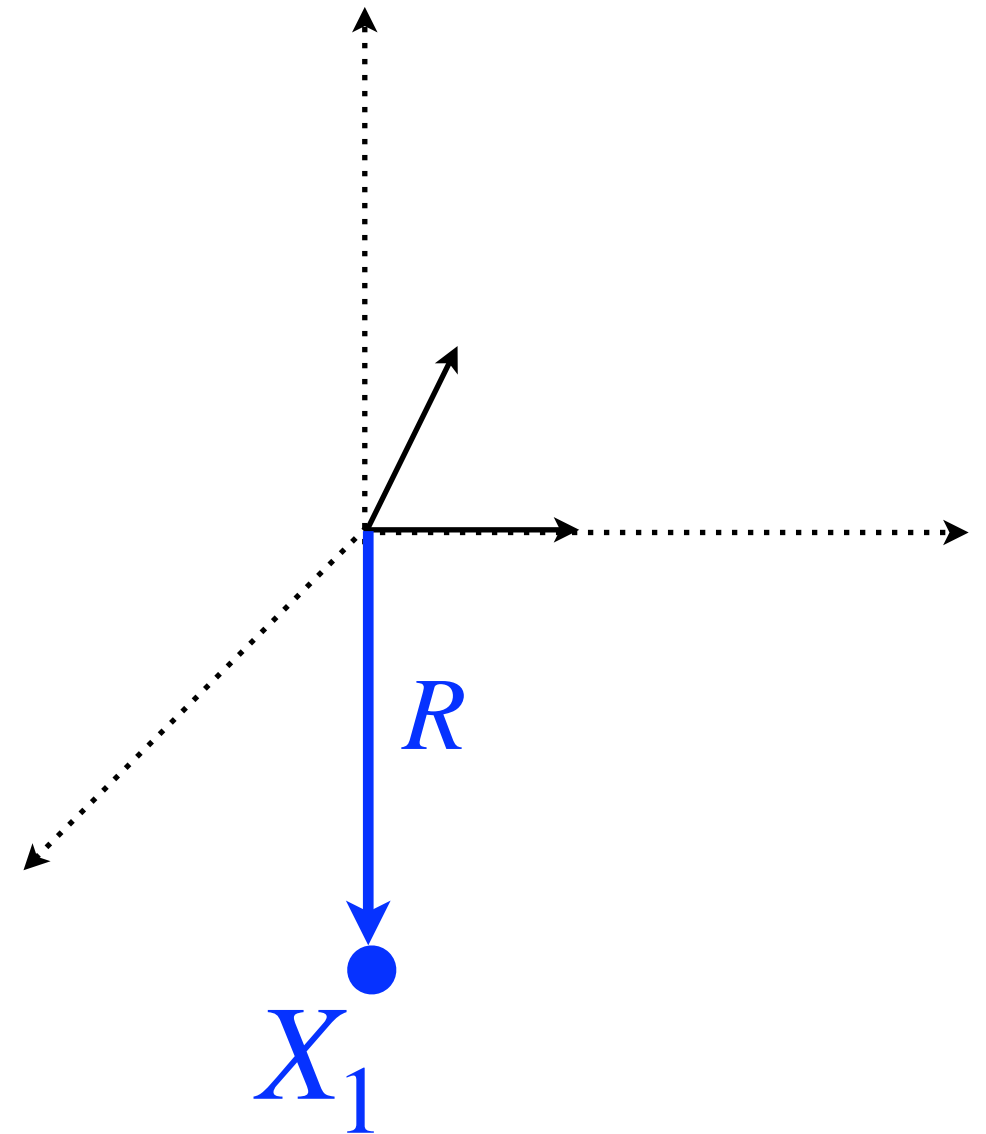
Exactly d directions is a simplifying assumption for the talk, can generalize with additional conditions

- Fixed creative directions $u_1, \dots, u_d$ (unit vectors) spanning the creative space $\mathbb{R}^d$



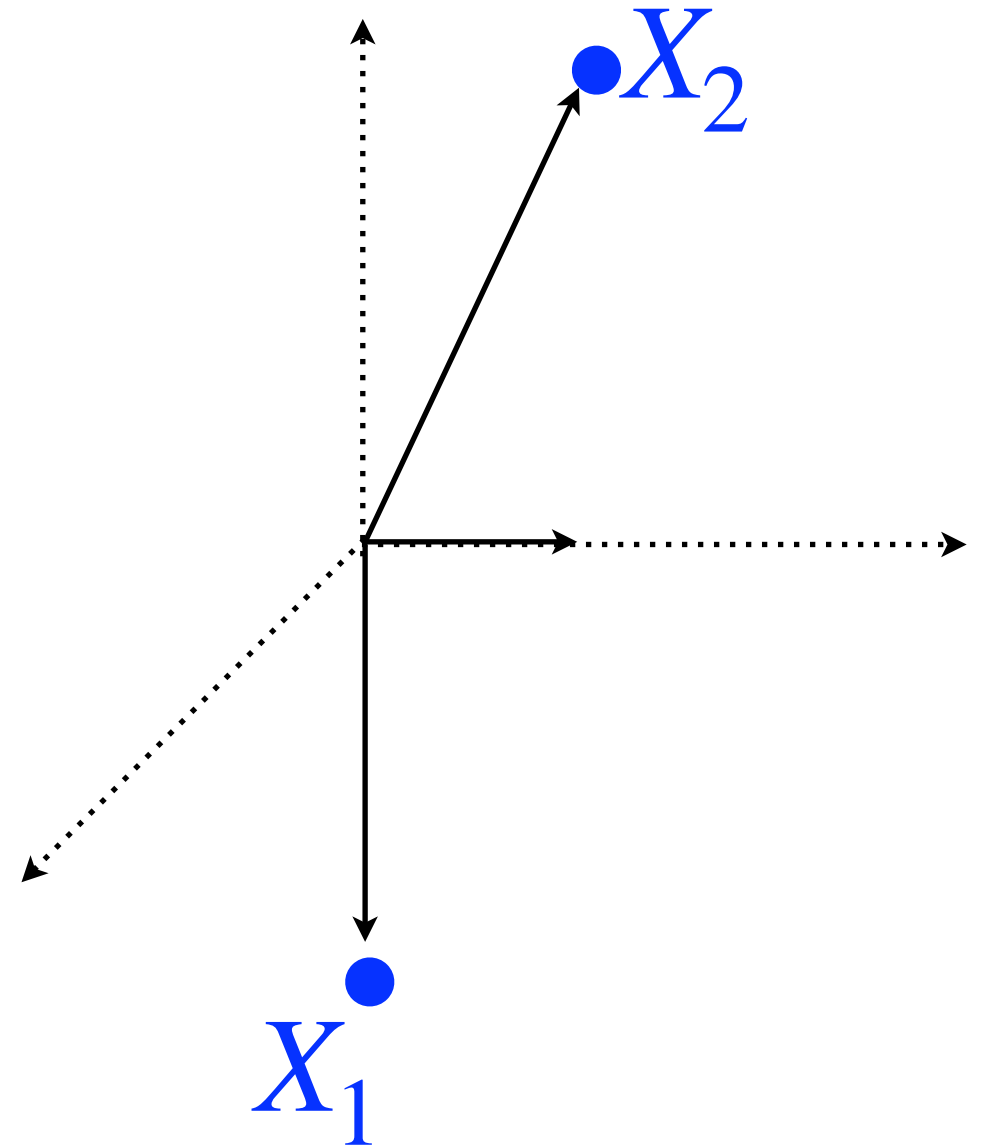
Corpus Formation

- Fixed creative directions $u_1, \dots, u_d$ (unit vectors) spanning the creative space $\mathbb{R}^d$
- At each date n :
 - Choose a direction $k \in \{1, \dots, d\}$ (possibly given the history)
 - Draw a magnitude $R \in \mathbb{R}_+$ from the distribution F_k
- The new creation is $X_n = Ru_k$



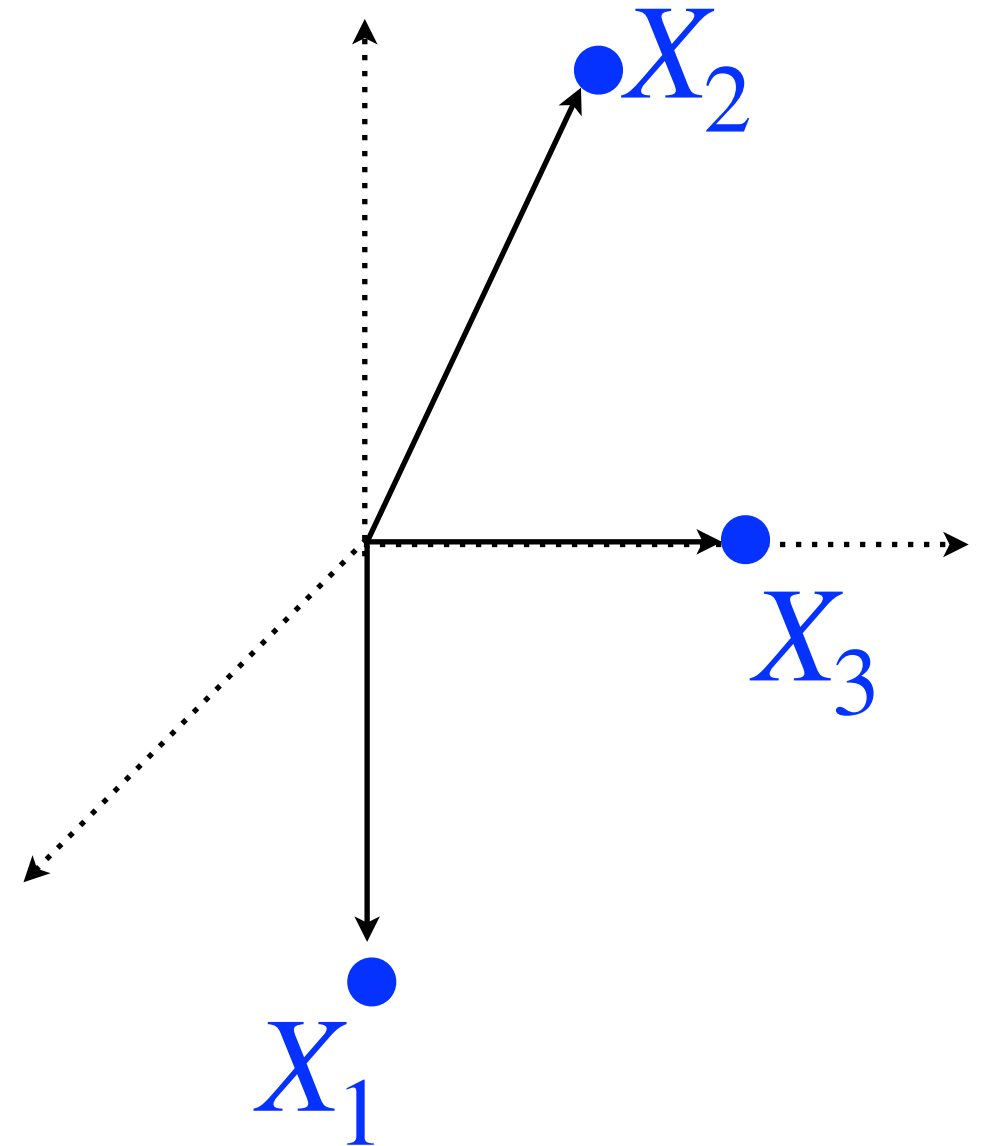
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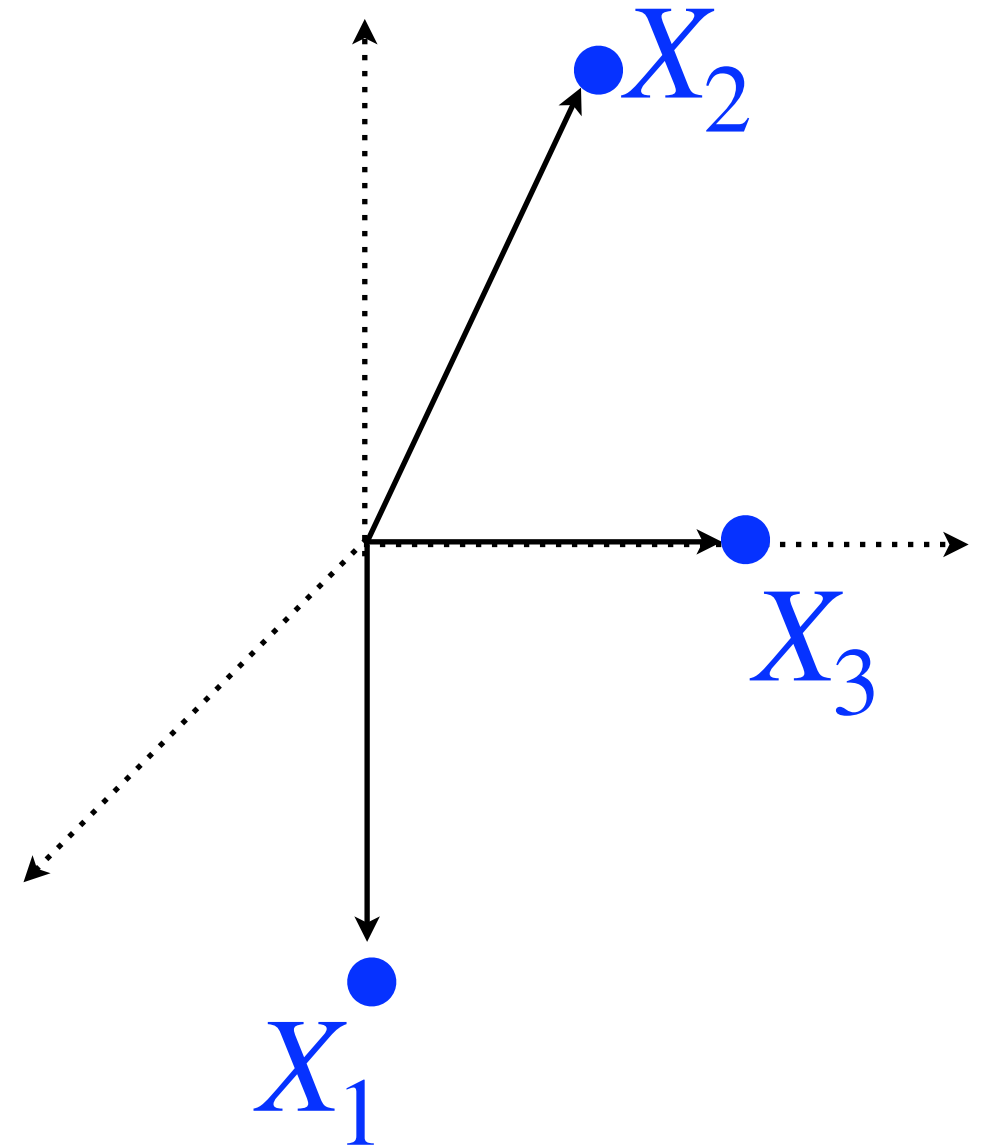
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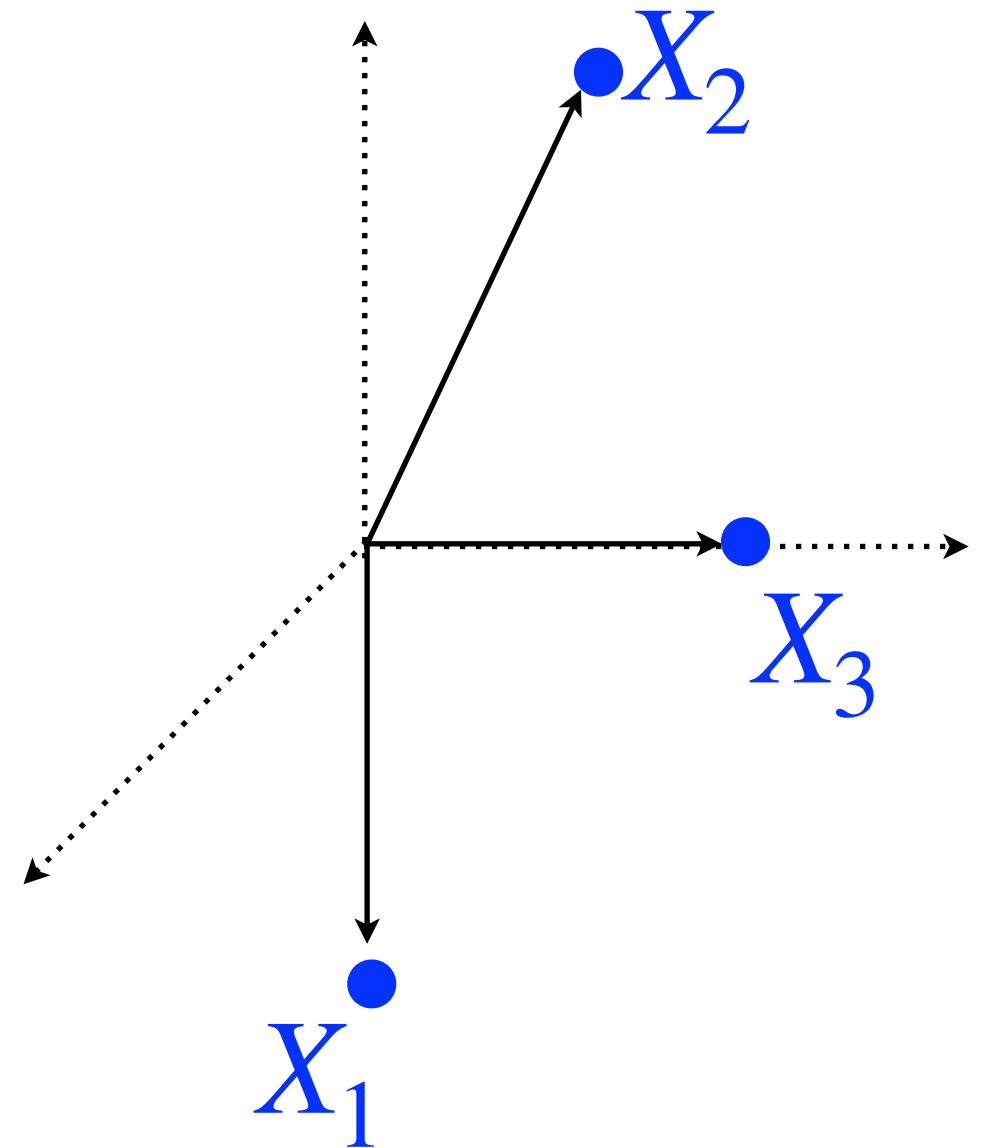
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Corpus Formation

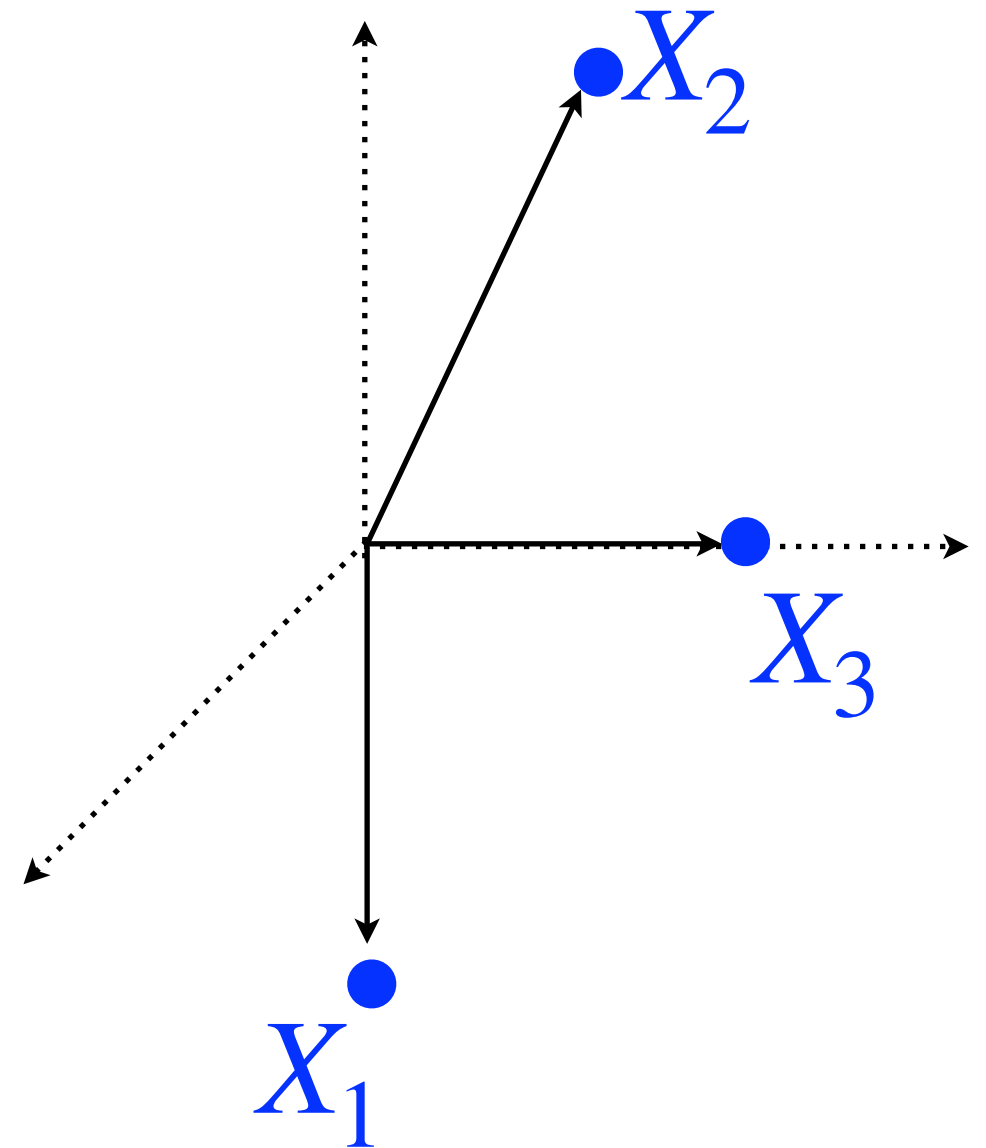
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Question: What is $\lim_{n \rightarrow \infty} r_g(C_n)$, i.e. the fraction of generable works that are eventually permissible?

Corpus Formation

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Main Result

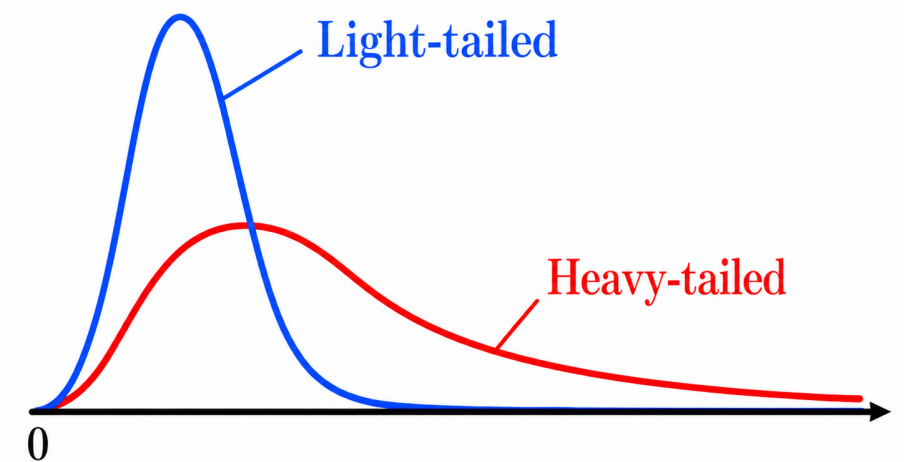
Theorem (informal): Fix a convex-valued generator g and suppose each direction is chosen in a positive fraction of periods.

A. If each F_k has light tails, then

$$\lim_{n \rightarrow \infty} r_g(C_n) = 1 \text{ a.s.}$$

B. If at least one F_k is heavy-tailed, then:

$$\liminf_{n \rightarrow \infty} r_g(C_n) = 0 \text{ a.s.}$$



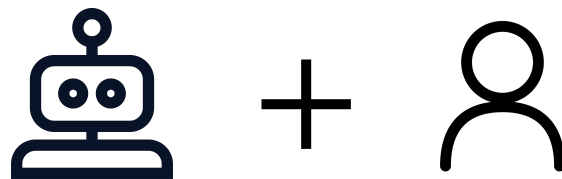
Takeaways

- In creative markets where innovation is light-tailed, individual creators are eventually marginal
- If there are rare but significant breakthroughs,
 - individual creators remain pivotal
 - our criterion strongly binds even with arbitrarily large corpora
- Future direction: which regime might we expect to emerge endogenously?
- e.g., incentives for creators might push towards a heavy-tailed regime (especially if creative protections follow our proposal)

AI and the Production of Research

Annie Liang
(Northwestern)

Tylor Wu
(Northwestern)



Introduction

- Researchers are increasingly using AI to do research. How will that affect the quality of research?
- AI reliance could reduce the quality of individual papers, e.g., might crowd out researcher effort or generate mistakes that go undetected

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- Abstract away from these concerns and suppose that AI makes papers both **less costly to produce** and also (up to a point) **higher in quality**
- Focus on a single downside: AI-assisted papers are more positively correlated with one another
 - e.g., Doshi and Hauser (2024) find that AI-assisted writers produce individually stronger but more collectively similar stories
 - see also Jiang et al 25, Ballesterro, Hosseini, Khanna, and Shorrer 26

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Question: How costly is this correlation?

Model

There are n researchers. Each researcher i chooses an **AI-reliance parameter** $\alpha_i \in [0,1]$ and a **number** of papers $m_i \in \mathbb{Z}_+$.

Paper qualities are modeled as jointly Gaussian random variables Y_{ik}

Higher reliance on the AI does three things:

- Papers get **less costly** to produce
- Papers (mostly) get **better**—expected quality rises up to an interior peak $\bar{\alpha}$, then falls
- Papers get **more alike**—correlation of paper qualities of researchers i and j is increasing in their AI reliance

Researcher's Payoff Function

Each researcher i

- earns **quality-based rewards** for the papers they produce
- incurs a **cost** for producing those papers
- shares in **scientific progress** (the best paper across researchers)

$$\sum_{k=1}^{m_i} \underbrace{\mathbb{E}[r(Y_{ik})]}_{\text{quality-based rewards}} - c_{\alpha_i}(m_i) + \gamma \mathbb{E} [\max \{ Y_{ik} \}_{i,k}]$$

the reward function r is
strictly increasing

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the cost function is pointwise
decreasing in α_i (AI makes
generation less costly)

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$\gamma \geq 0$ is the weight placed on
the social value of research

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PRIVATE VALUE OF
RESEARCH

SOCIAL VALUE OF
RESEARCH

Equilibrium and Welfare

Consider two games:

- The **AI Game** Γ_n^{AI} in which each of n researchers chooses (α, m)
- The **pre-AI Game** Γ_n^0 in which AI reliance is fixed at $\alpha = 0$ for all researchers, so they choose only m

Our solution concept is Nash equilibrium.

Welfare $W(a)$ is the **sum of player utilities** at the action profile

$$a = ((\alpha_1, m_1), \dots, (\alpha_n, m_n))$$

Main Result

Theorem: Fix sequences (a_n^{AI}) and (a_n^0) where each a_n^{AI} is a NE of the AI game Γ_n^{AI} and each a_n^0 is a NE of the pre-AI game Γ_n^0 . Then

- If $\gamma = 0$, then $W(a_n^{AI}) > W(a_n^0)$ for all n
- If $\gamma > 0$, then $W(a_n^{AI}) < W(a_n^0)$ for all n sufficiently large (and the gap grows without bound)

In words:

- if **no** weight is put on the social value of research ($\gamma = 0$), then welfare is higher with AI
- if **any** weight is put on the social value of research ($\gamma > 0$), then welfare is higher without the option of AI

Discussion

$$\sum_{k=1}^{m_i} \mathbb{E}[r(Y_{ik})] - c_{\alpha_i}(m_i) + \gamma \mathbb{E} [\max \{ Y_{ik} \}_{i,k}]$$

- When $\gamma = 0$, private incentives and welfare are perfectly aligned, and AI expands the researcher's choice set
- With access to AI, researchers generate **more** papers and **better** papers, but these are correlated
- The latter effect wins out

Interpretation

- This result is a cautionary tale about AI use
- I don't take this as saying that AI is going to impede scientific progress or welfare
- Rather, it helps us to clarify the assumptions that need to be broken for certain outcomes to hold

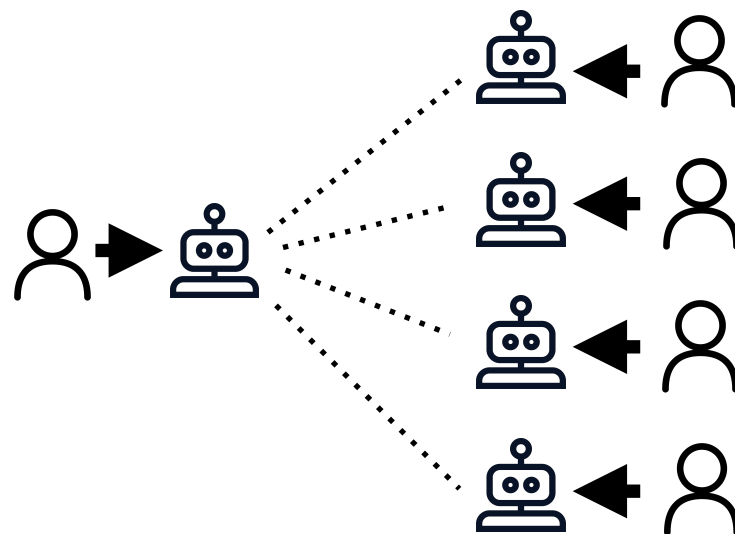
Extensions and Limitations

Some results in progress:

- The main result **generalizes** if the social value of research is not simply the best paper, but a weighted average of the top K
- The main result **flips** if we assume that research quality is heavy-tailed (regularly varying)

Artificial Intelligence Clones

Annie Liang
(Northwestern)



REPLICANT

AI-powered candidate replication platform helping job seekers create intelligent digital representatives for recruitment conversations.



AI Training

Upload your resume and provide context about your experience, goals, and preferences



Clone Creation

Our AI creates an intelligent representation that understands your professional profile



Smart Matching

Your AI clone engages with recruiters, answering questions and showcasing your skills

Ready to Get Started?

Join the future of recruitment and let your AI clone work for you.

Create Your AI Clone

Ribbon raised \$8.2M to fix hiring globally >

Meet your AI-powered recruiter

Ribbon talks with every candidate, scores their answers, and shares insights that help you hire faster and smarter.

Book a demo

Start for free



Daphne D.
Operations Manager

This was my first time doing an interview like this and it was so smooth!



Hassan J.
DevOps Engineer

I like the new AI approach, and I thought the agent was very interactive and smart and responded well to changing dynamics of the conversation!



Jamie K.

Are A.I. Clones the Future of Dating? I Tried Them for Myself.

executives in the dating app industry have begun pitching a future in which people can create A.I. clones of themselves that date other clones and relay the results back to their human counterparts.

Dating Apps Suck. A.I. Clones Are Making Them Even Weirder.

‘My clone is on its way to date all of San Francisco in one month.’

Setup

Individuals are represented by vectors in $\mathbb{R}^k$

- each entry is an attribute of the person

possible attributes:

extraversion

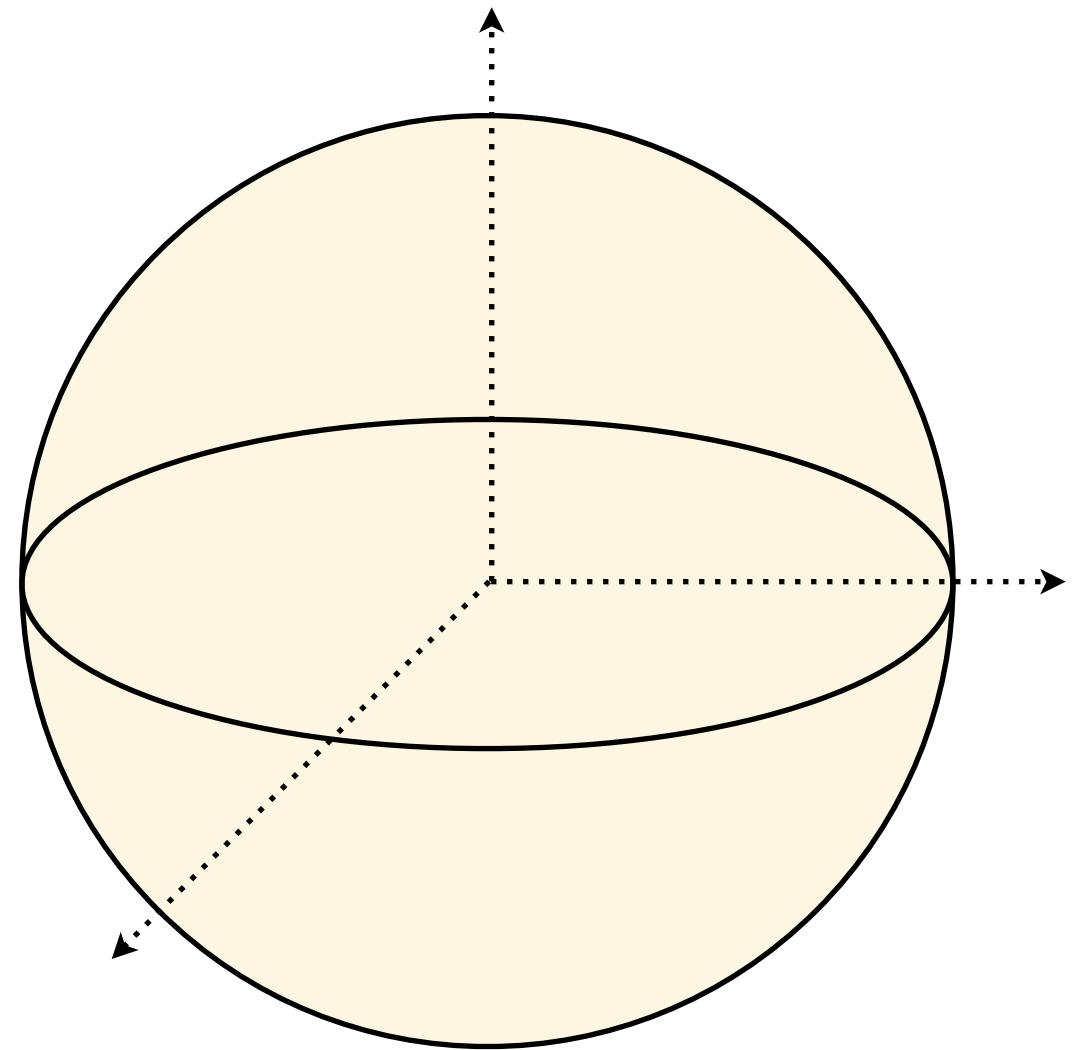
intellectual curiosity

relationship to time

style of humor

relationship to authority

speed of decision-making

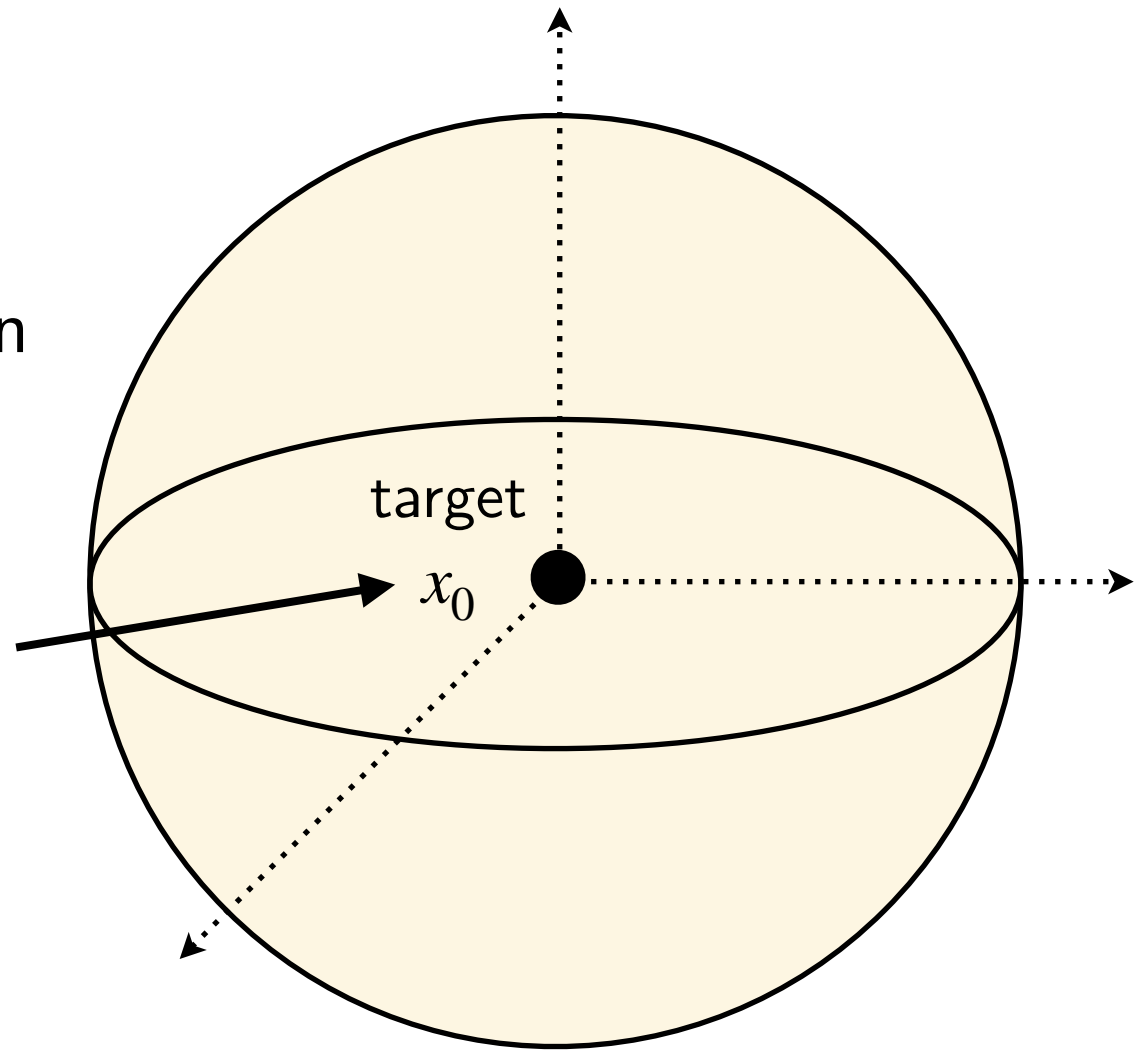


Setup

Individuals are represented by vectors in $\mathbb{R}^k$

- each entry is an attribute of the person

The searcher's ideal match is normalized to the origin and denoted x_0 (the **target**)



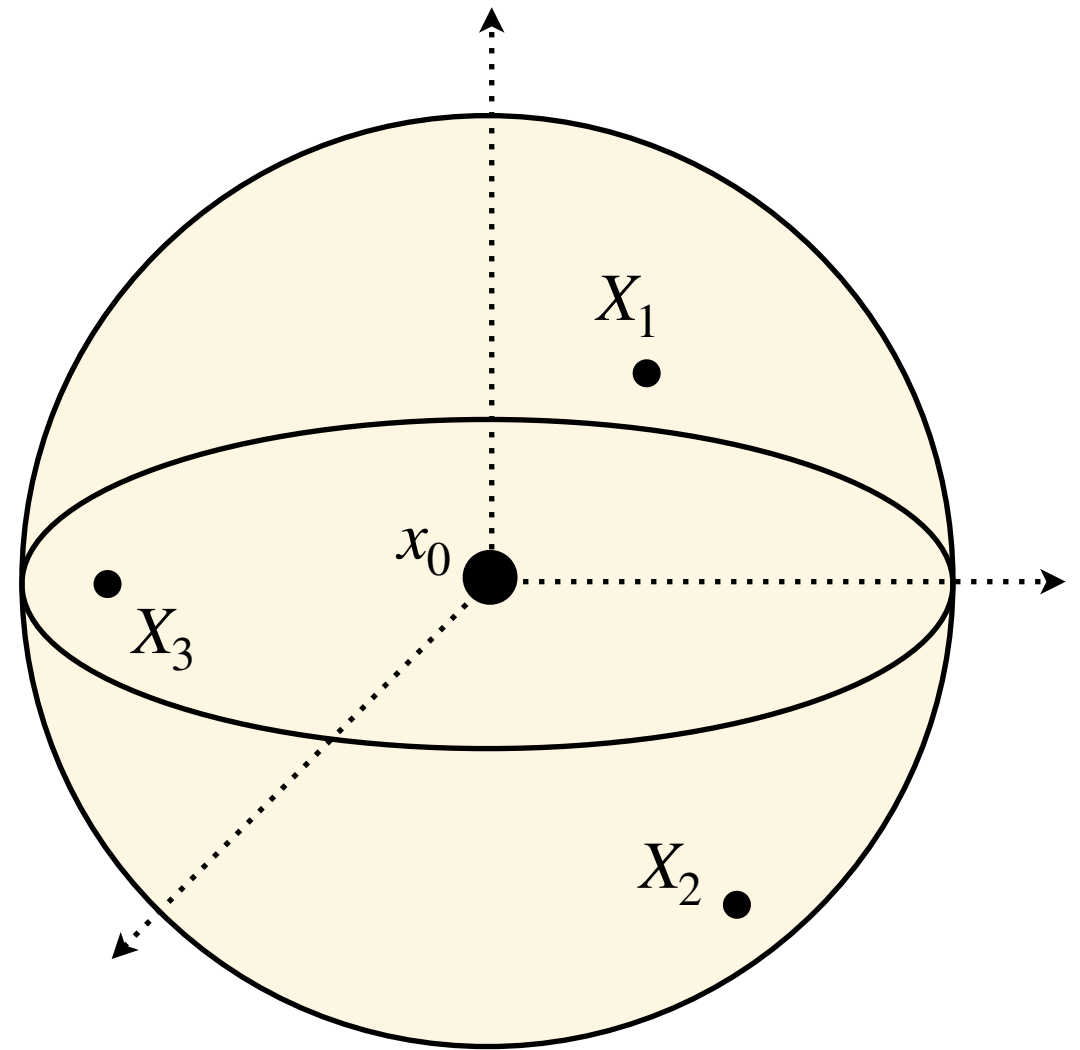
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People in the population $X_i \in \mathbb{R}^k$ are drawn uniformly at random from the unit ball B_k



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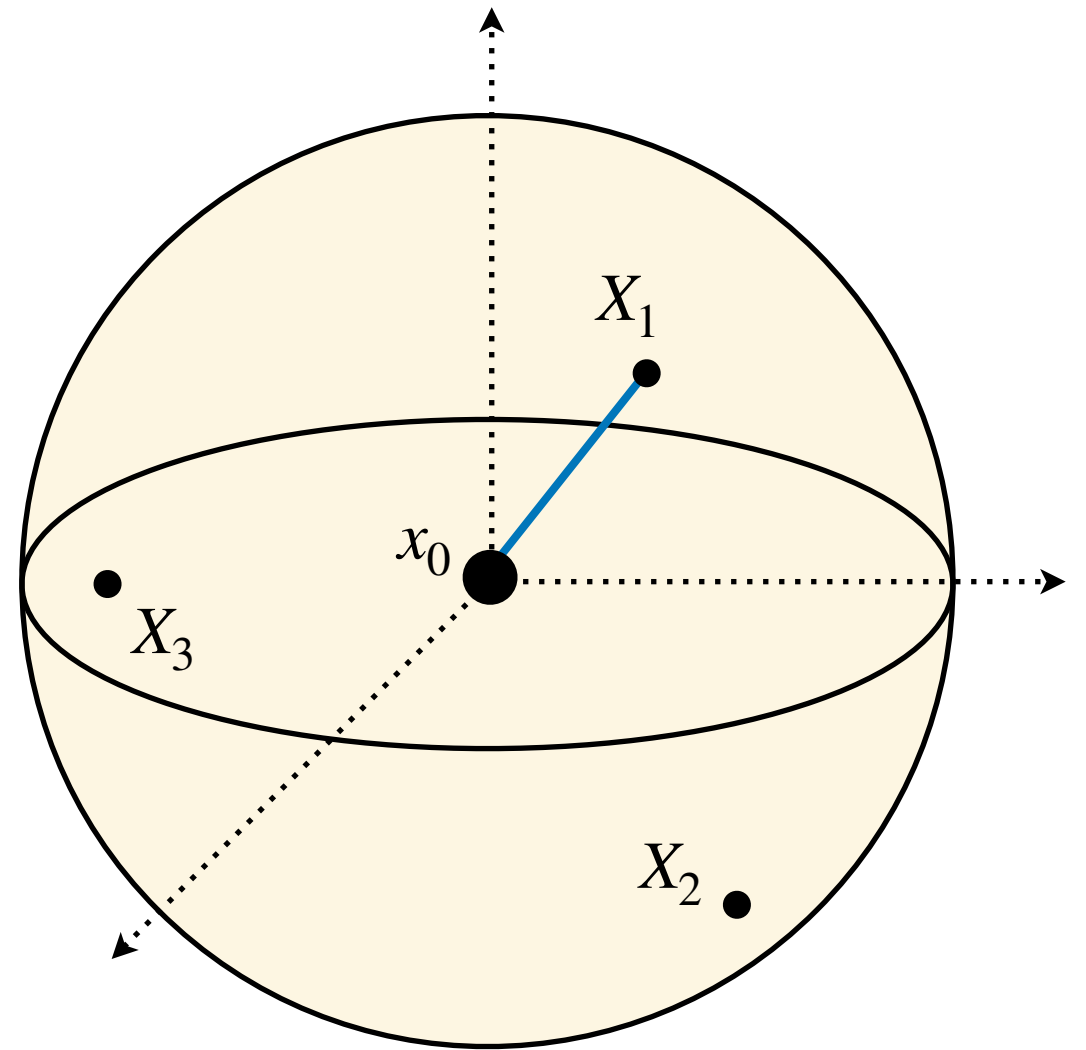
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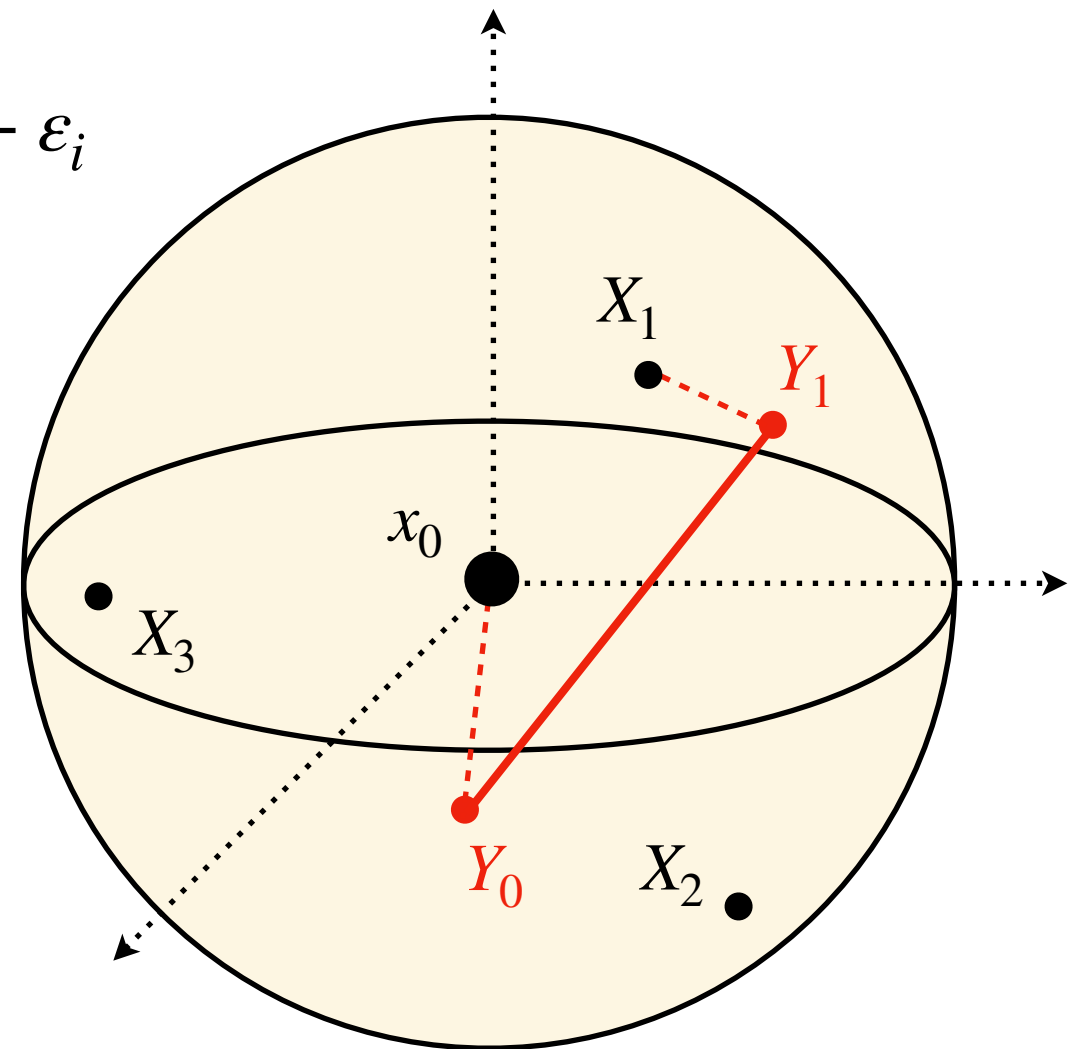
The searcher receives utility $-\|X_i\|$ from being matched to individual i



AI-Mediated Search

Under **AI-mediated** search,

- The searcher's target is represented by $Y_0 = x_0 + \varepsilon_0$
- Each individual i is represented by $Y_i = X_i + \varepsilon_i$
- The noise terms $\varepsilon_0, \varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_k)$ are independent of one another
- Henceforth refer to Y_i and Y_0 as *AI representations*

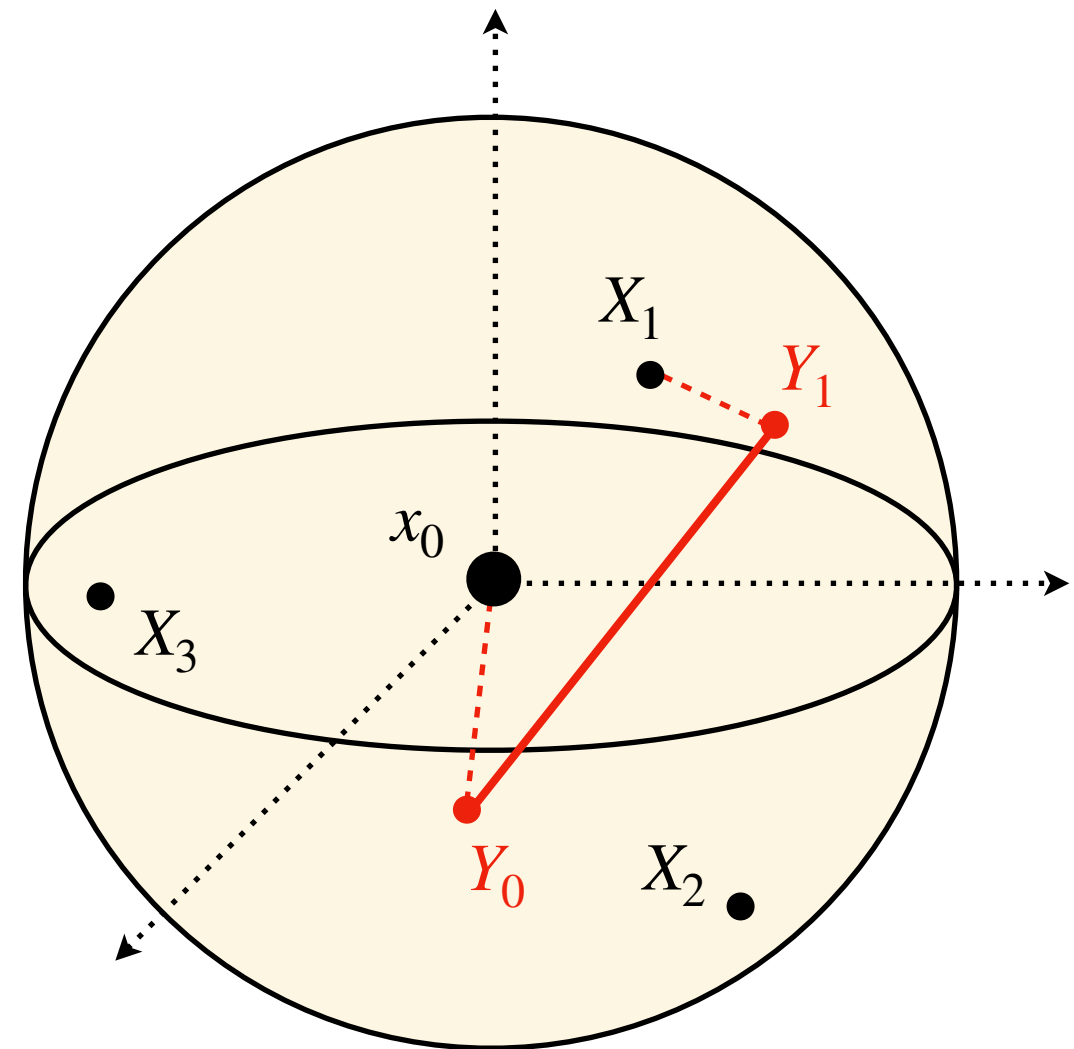


AI-Mediated Search

- The AI platform searches over n possible matches for the searcher
- The individual with the most compatible AI representation is

$$i_n^* = \operatorname{argmin}_{1 \leq i \leq n} \underbrace{\|Y_i - Y_0\|}_{\text{distance between the representation of the ideal point and the representation of individual } i}$$

distance between the representation
of the ideal point and the
representation of individual i



AI-Mediated Search

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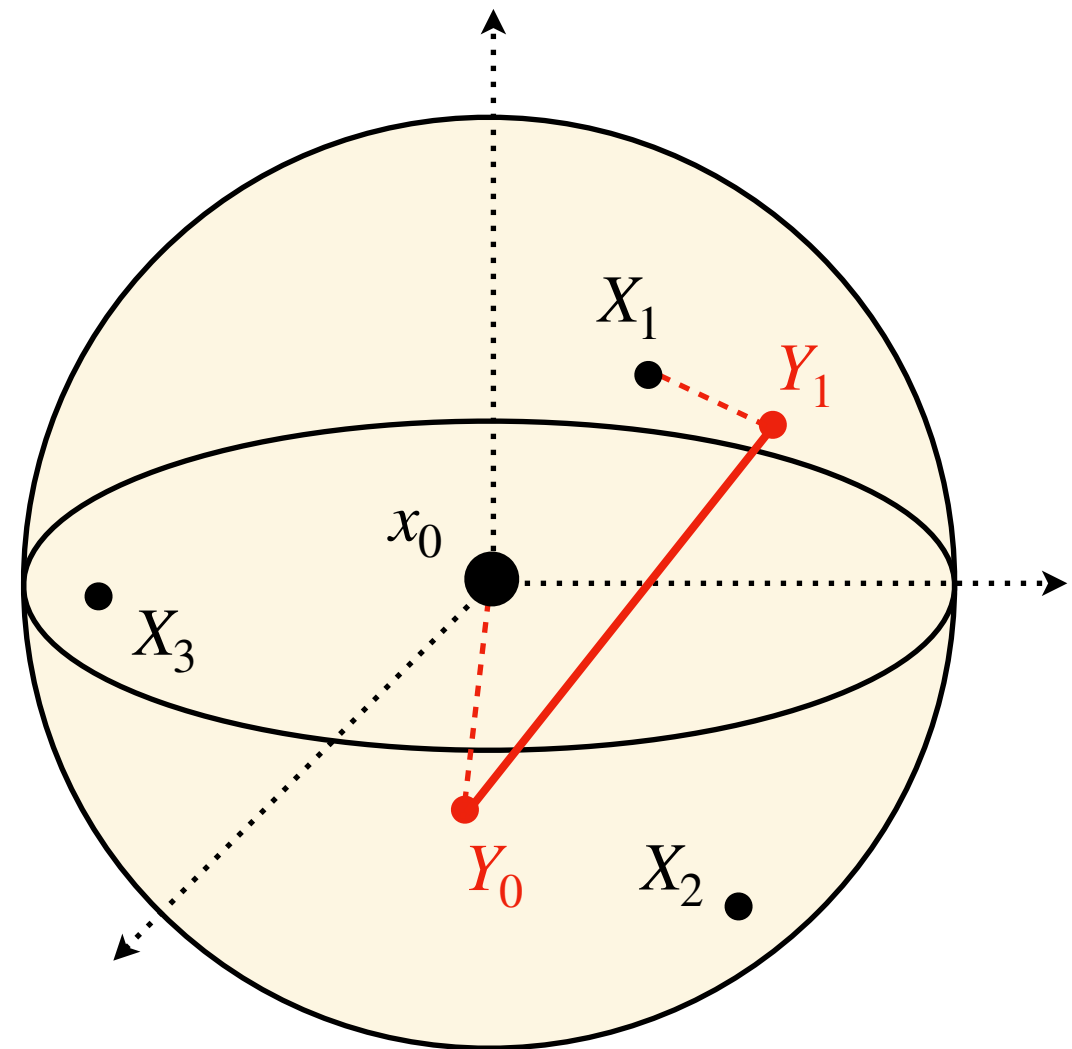
- The individual with the most compatible AI representation is

$$i_n^* = \operatorname{argmin}_{1 \leq i \leq n} \|Y_i - Y_0\|$$

- The searcher's expected utility is

$$u_k^{AI}(n) \equiv -E \left[\|X_{i_n^*}\| \right]$$

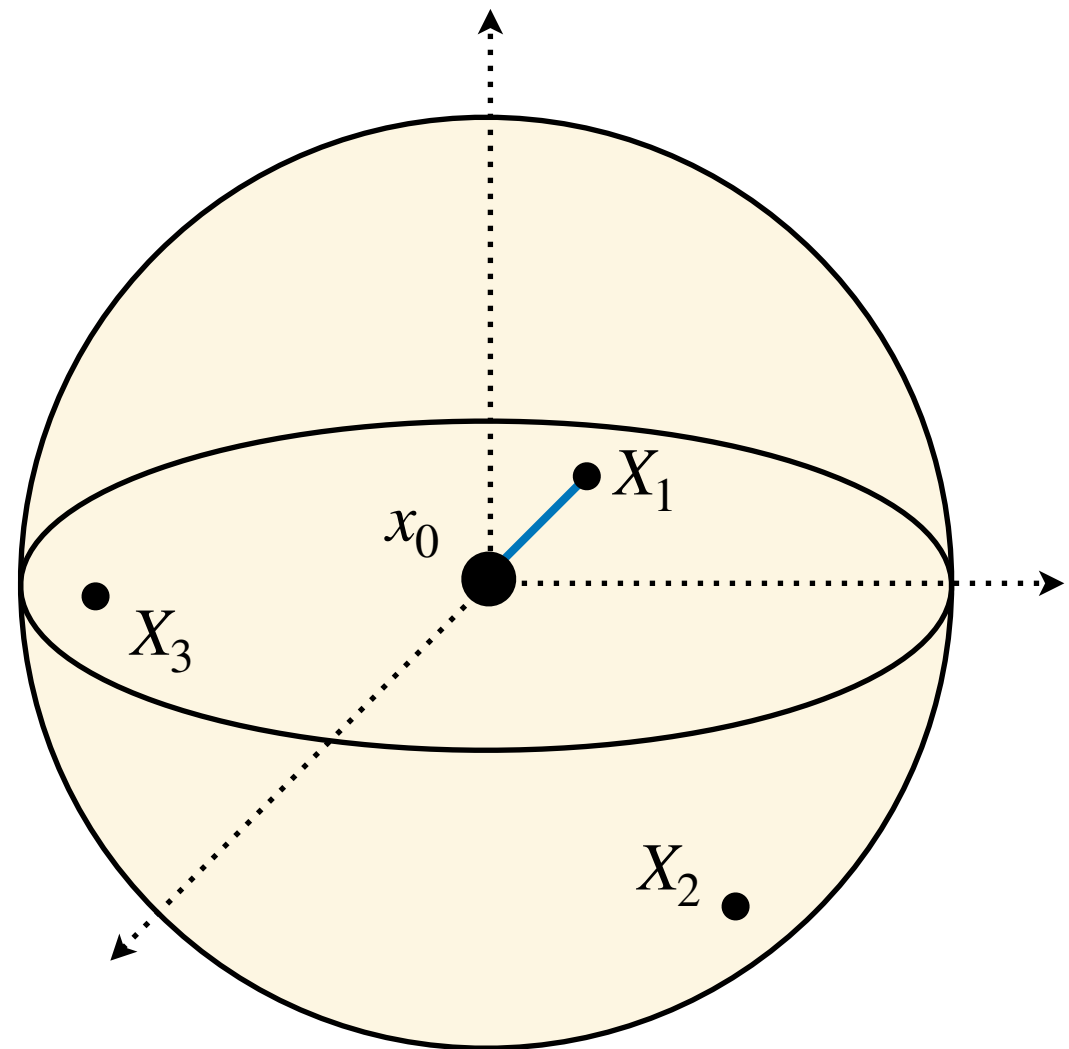
The selected match minimizes the distance between representations, but the searcher's payoff is based on the **true** distance between the underlying points



Benchmark: Best among m

- Suppose the searcher were to meet m individuals $X_1, \dots, X_m$ and match to the truly most compatible among them
- The searcher's expected utility from matching to this "best among m " is

$$u_k^B(m) \equiv - \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$$



AI-Equivalent Sample Size

Measure of the value of AI-mediated search:

Definition: For any number of dimensions k , the *AI-equivalent sample size* m_k^* is the smallest $m \in \mathbb{Z}_+$ satisfying

$$u_k^B(m) > u_k^{AI}(n) \quad \forall n \in \mathbb{Z}_+$$

if the inequality above is not satisfied for any finite m , then $m_k^* = \infty$.

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expected utility of best match
among m people

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expected utility of best match
on platform with n candidates
(AI-mediated search)

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$$\underbrace{u_k^B(m) > u_k^{AI}(n)}_{\text{inequality}} \quad \forall n \in \mathbb{Z}_+$$

if the inequality above is not satisfied for any finite m , then $m_k^* = \infty$.

matching to the best candidate among m draws beats the AI platform, no matter how large it is

Main Results

Proposition: For every number of dimensions $k \in \mathbb{Z}_+$, the AI-equivalent sample size m_k^* is finite

- The value to increasing scale is capped

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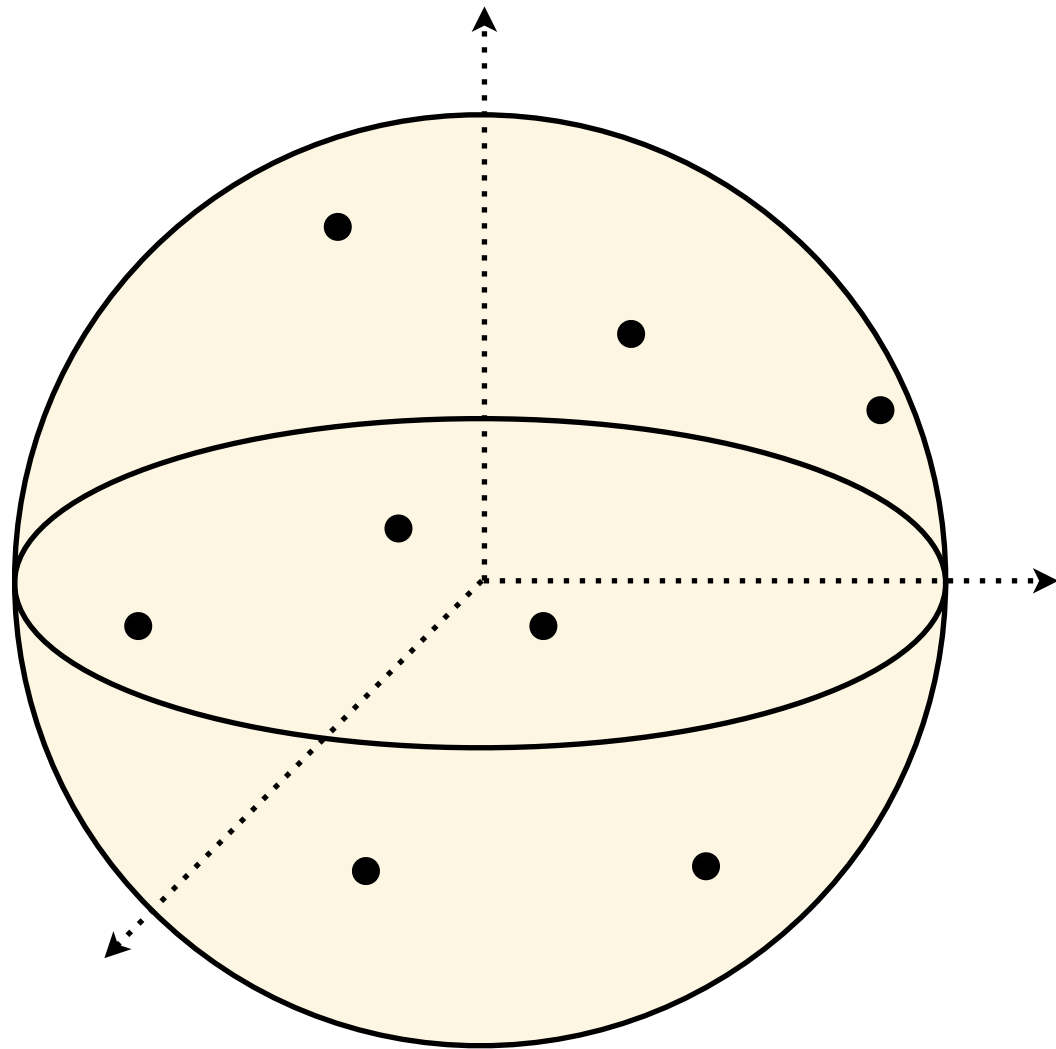
Theorem: For all k sufficiently large, the AI-equivalent sample size is $m_k^* = 2$

- When there are sufficiently many relevant attributes, being on the platform is no better than meeting two people and matching to the better among them
- Underscores the limits of AI representations for complex search problems

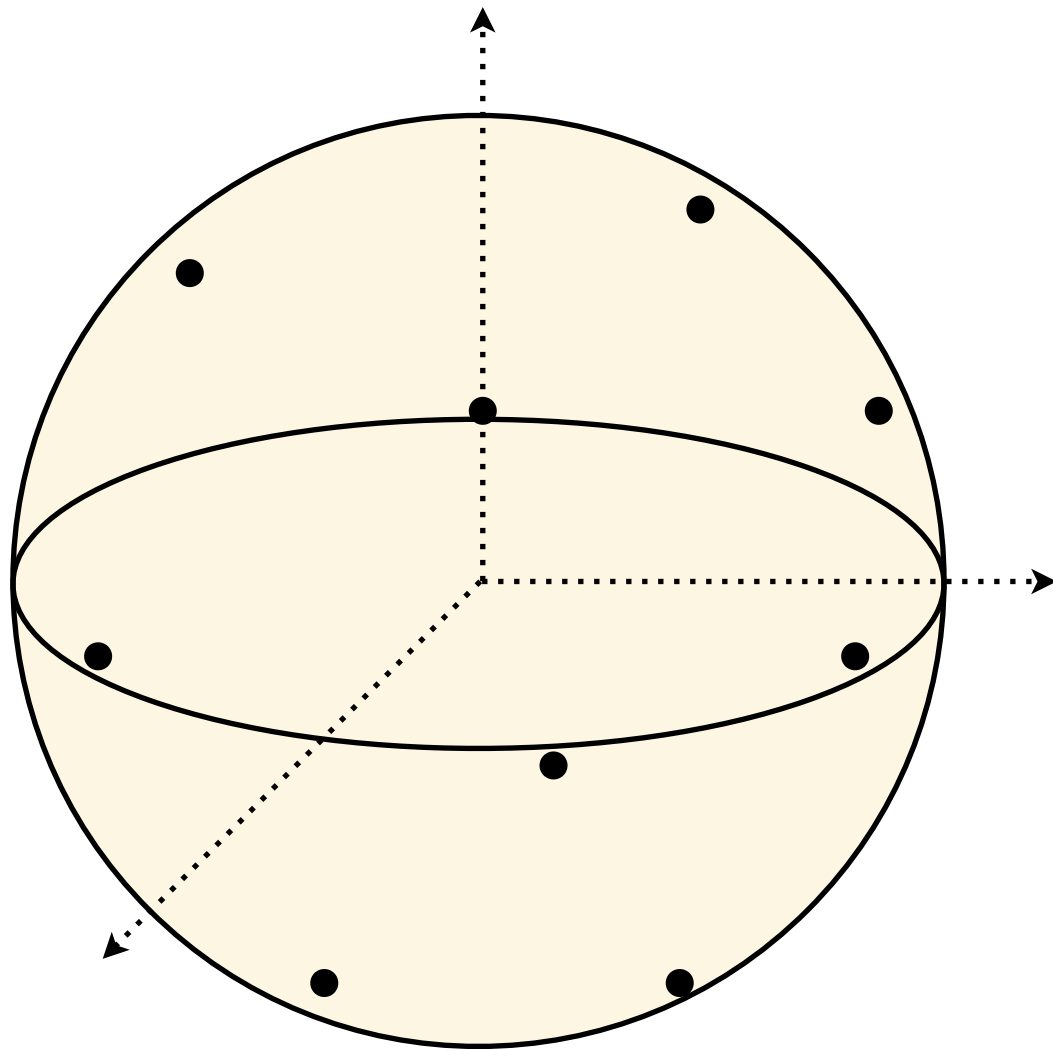
How Large is k ?

- In a simulation with $\sigma = 0.05$, the result holds for $k = 150$ dimensions
- How many dimensions of personality are relevant for identifying
 - good spousal matches?
 - good employees?
- This is an empirical question
- The main takeaway is that **the complexity of match compatibility matters** for how we should think about human vs. AI search
- The value of AI representations can differ across search problems depending on what the relevant k is

Proof Idea/Intuition



Proof Idea/Intuition

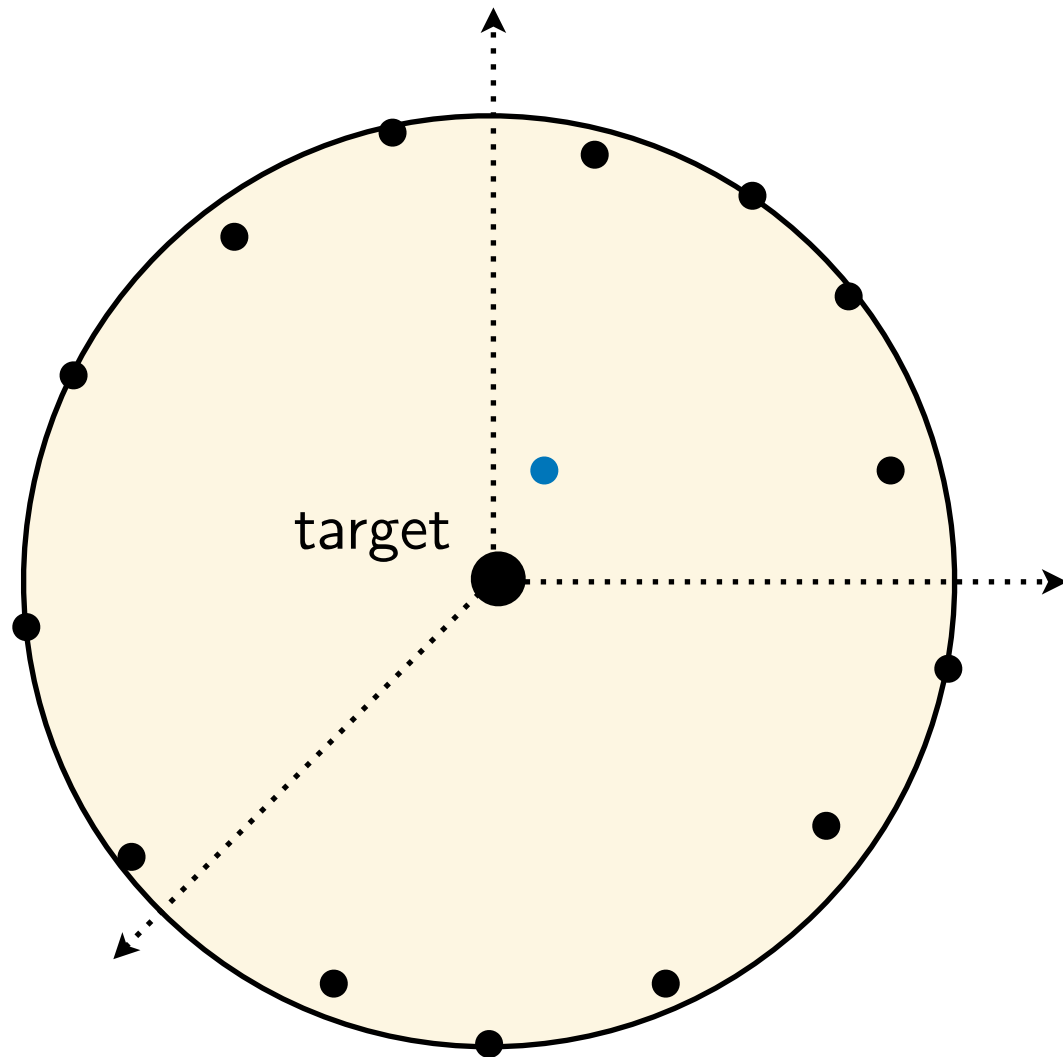


In high dimensions:

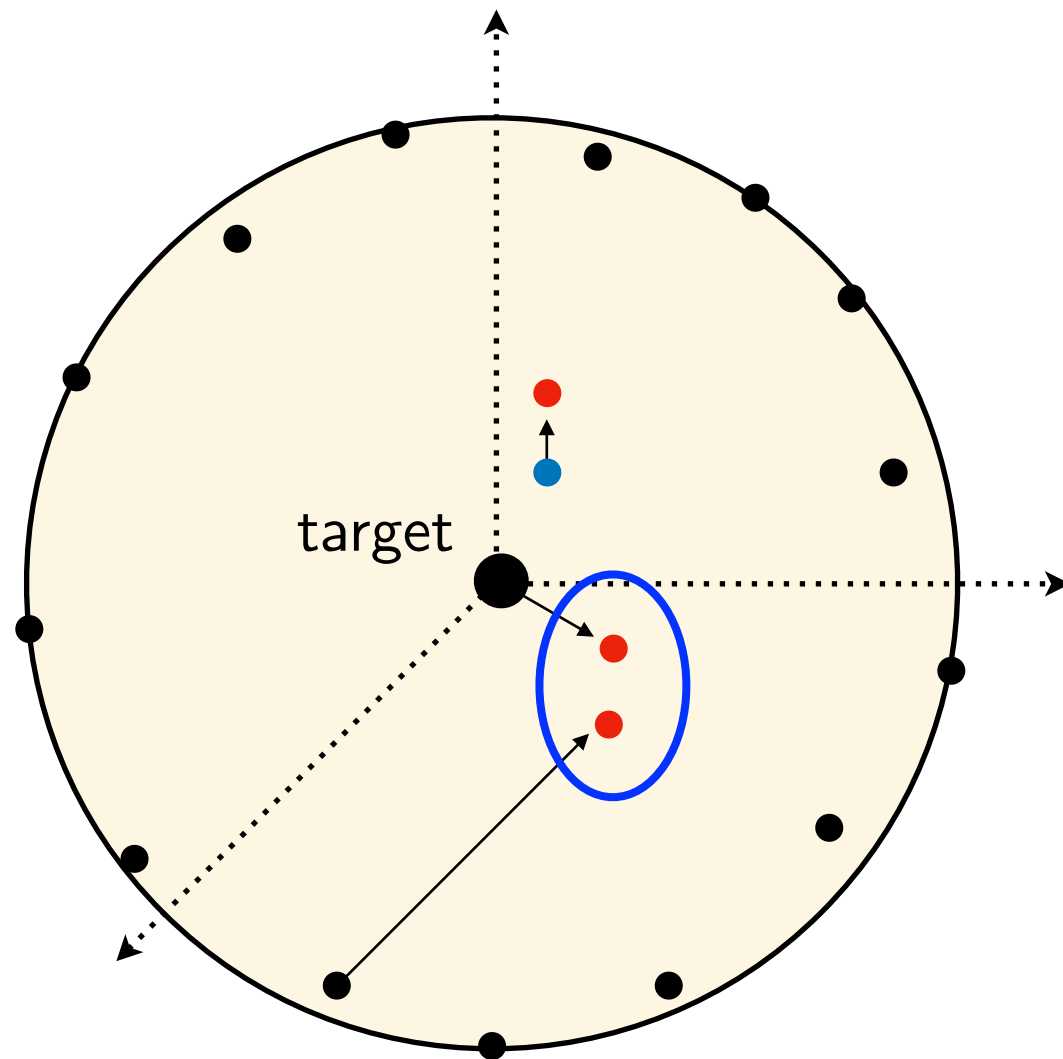
- points amass on the boundary of the unit ball
- distances between them concentrate around their expectations

i.e., people are increasingly unique and increasingly unlikely to exactly match the target

Proof Idea/Intuition

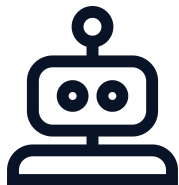


Proof Idea/Intuition

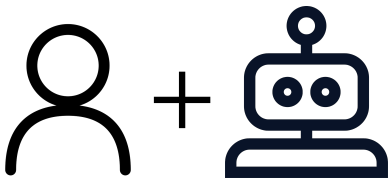


- Best match increasingly driven by realization of the noise terms
- Eventually as if just selecting one candidate at random
- So matching to the best of two candidates is better

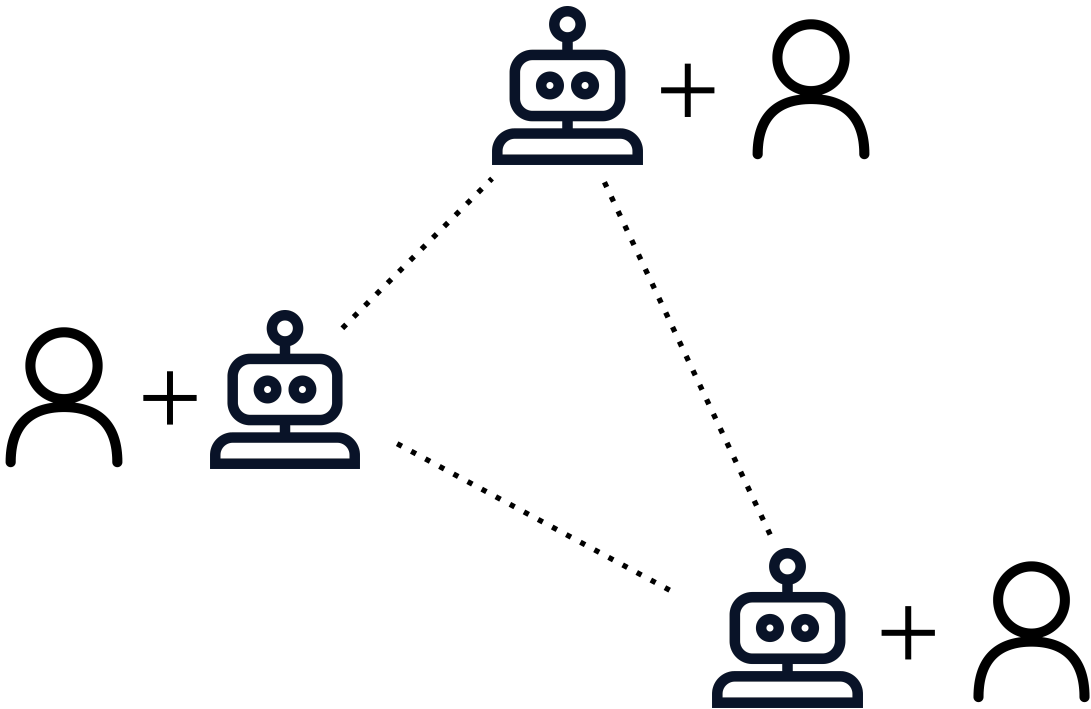
Conclusion



Gao, Han, and
Liang (2026)



Liang and Lu
(2026)
Liang and Wu
(2026)



Liang (2025)